PEARSON EDEXCEL GCSE (9–1) HISTORY · PAPER 1 (1HI0/11)

Section A 12-Page Pack

The British Sector of the Western Front, 1914–1918

The Historic Environment · Injuries, Treatment and the Trenches · Complete Mastery & Source Studio

Pupil Name: ______________________ **Class / Set:** __________ **Target Grade:** [ ]

📊 SECTION A PROGRESS & ASSESSMENT TRACKER (16 MARKS TOTAL)

Marked by Teacher or Peer Verified

Assessment Component & Stem Focus	Format Style	Max Marks	Pupil Score
🧠 **Complete Western Front Knowledge Vault (All 100 Recall Questions)**	**Checklist**	**/100**	_____
Round 1 (Stepped): Q1(a) Feature 1 (Casualty Clearing Stations)	Stepped Ladder	/2	_____
Round 1 (Stepped): Q1(b) Feature 2 (Thomas Splint)	Stepped Ladder	/2	_____
Round 1 (Stepped): Q2(b) 4-Part Follow-Up Grid (Base Hospital Water)	Stepped Ladder	/4	_____
Round 1 (Stepped): Q2(a) Source Utility (Somme CCS vs Base Hospital)	Stepped Ladder	/8	_____
Round 2 (Dual-Track): Q1(a) & Q1(b) Features (Mobile X-rays & Trench Fever)	Dual Track	/4	_____
Round 2 (Dual-Track): Q2(b) 4-Part Follow-Up Grid (Gas Casualties)	Dual Track	/4	_____
Round 2 (Dual-Track): Q2(a) Source Utility (RAMC Logbook & Flanders Mud)	Dual Track	/8	_____
Round 3 (Simulation): Full Section A Timed Exam Pitch (Q1a, Q1b, Q2a, Q2b)	Exam Hall (25m)	/16	_____
TOTAL COMBINED EXAM PRACTICE MARKS:		**/48**	**_____ / 48**

Teacher / Peer Marker: ______________________ **Date:** __________ **Overall Grade: [9 8 7 6 5 4]**

⏱ SECTION A EDEXCEL EXAM FAST FACTS & STEM BLUEPRINTS (16 MARKS · ~25 MINS)

OPTION 11 PAPER 1 SECTION A

- **Q1(a) & Q1(b) Features (2 Marks each · ~3 mins each):** *F-D Formula.* State **Feature** in sentence 1 → support with **1 precise supporting factual detail**. (2 separate questions).
- **Q2(a) Source Utility (8 Marks · ~12 mins):** *C-O-P Matrix.* Evaluate both sources for **Content** (what it says) → **Own Contextual Knowledge** → **Provenance** (Nature, Origin, Purpose / NOP).
- **Q2(b) Follow-Up Investigation (4 Marks · ~7 mins):** *4-Step Enquiry.* Quote **Detail in Source B** → Formulate **Targeted Question** → Name **Specific Historical Source Type** → Explain **Direct Purpose**.

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Complete Knowledge Retrieval Vault · Questions 1 to 50 (Trench Environment, Ill-Health & Wounds)

VAULT PART 1

- [ ] **1.** By 1900, how had surgical practice in civilian hospitals evolved from Joseph Lister's earlier methods? [KT5.1]
- [ ] **2.** Which of the following was a key piece of technology used in aseptic surgery by the outbreak of the First World War? [KT5.1]
- [ ] **3.** In what year did the German physicist Wilhelm Roentgen accidentally discover X-rays? [KT5.1]
- [ ] **4.** Which of the following was a major limitation of using early X-ray machines in 1914? [KT5.1]
- [ ] **5.** Why were early X-ray machines highly unsuitable for the fast-paced, mobile environment of a battlefield in 1914? [KT5.1]
- [ ] **6.** In 1901, the Austrian doctor Karl Landsteiner made a breakthrough that revolutionized blood transfusions. What did he discover? [KT5.1]
- [ ] **7.** Before the discovery of sodium citrate in 1915, what was the massive problem with performing blood transfusions on the battlefield? [KT5.1]
- [ ] **8.** When the British Expeditionary Force (BEF) arrived on the Western Front in 1914, what geographical feature made the medical situation catastrophic? [KT5.1]
- [ ] **9.** What is the military definition of a 'salient', such as the famous one at Ypres? [KT5.1]
- [ ] **10.** Which major British offensive, launched in 1916, became a medical disaster with over 57,000 casualties on the very first day? [KT5.1]
- [ ] **11.** What unique medical facility was established during the Battle of Arras in 1917? [KT5.1]
- [ ] **12.** During the Battle of Cambrai in 1917, what groundbreaking medical logistical system was used for the first time by Oswald Robertson? [KT5.1]
- [ ] **13.** Why did aseptic surgery (sterile operating theatres) prove to be nearly useless in frontline dressing stations in 1914? [KT5.1]
- [ ] **14.** What made transporting wounded soldiers away from the frontline so incredibly difficult? [KT5.1]
- [ ] **15.** To what extent was the Royal Army Medical Corps (RAMC) prepared for the medical challenges of the Western Front in 1914? [KT5.1]
- [ ] **16.** What did the term 'war of attrition' mean for the soldiers living in the trench system? [KT5.1]
- [ ] **17.** Why were horse-drawn ambulances often ineffective in the worst conditions of the Western Front? [KT5.1]
- [ ] **18.** Which of the following best describes the layout of the trench system on the Western Front? [KT5.1]
- [ ] **19.** In what year did Karl Landsteiner discover that blood group O was the universal donor group? [KT5.1]
- [ ] **20.** Which statement accurately summarises the 'historical context' of medicine at the outbreak of war in 1914? [KT5.1]
- [ ] **21.** What was the direct environmental cause of 'Trench Foot' on the Western Front? [KT5.2]
- [ ] **22.** What were the symptoms of severe, untreated Trench Foot? [KT5.2]
- [ ] **23.** How did the British Army attempt to clinically prevent the spread of Trench Foot in permanently flooded trenches? [KT5.2]
- [ ] **24.** What was 'Trench Fever', and what were its primary symptoms? [KT5.2]
- [ ] **25.** It took years for medical officers to discover the cause of Trench Fever. What was eventually found to be the transmitter? [KT5.2]
- [ ] **26.** How did the British Army strategically adapt to combat the epidemic of Trench Fever once its cause was discovered? [KT5.2]
- [ ] **27.** What was the 'Ypres Salient'? [KT5.2]
- [ ] **28.** Why was the First Battle of Ypres (Autumn 1914) strategically critical for the British? [KT5.2]
- [ ] **29.** What terrifying new weapon was first used by the German Army at the Second Battle of Ypres in 1915? [KT5.2]
- [ ] **30.** During the fight for Hill 60 in April 1915, what tactic did the British successfully use to defeat the German positions? [KT5.2]
- [ ] **31.** Why did the British launch the massive Battle of the Somme in 1916? [KT5.2]
- [ ] **32.** What made the Battle of the Somme a 'medical nightmare' on its very first day? [KT5.2]
- [ ] **33.** Because the ground at Arras was chalky and easy to tunnel through, what incredible feat did British and New Zealand miners achieve in 1917? [KT5.2]
- [ ] **34.** In the organisation of the trench system, what was the primary purpose of the 'support trench', located roughly 80 metres behind the frontline? [KT5.2]
- [ ] **35.** Why were trenches deliberately dug in a zig-zag pattern instead of a straight line? [KT5.2]
- [ ] **36.** How did the complex, zig-zag design of the trenches unintentionally create severe problems for the medical services? [KT5.2]
- [ ] **37.** Why did the army have to rely heavily on horse-drawn ambulance wagons instead of modern motor ambulances near the frontline? [KT5.2]
- [ ] **38.** What was the main medical danger caused by delays in evacuating wounded soldiers across the devastated, muddy terrain? [KT5.2]
- [ ] **39.** What was 'Dysentery' and how was it spread on the Western Front? [KT5.2]
- [ ] **40.** What does the medical condition 'Shell Shock' (now understood as PTSD) reveal about the environment of the Western Front? [KT5.2]
- [ ] **41.** Which type of weapon was responsible for the highest percentage of wounds (roughly 58%) on the Western Front? [KT5.3]
- [ ] **42.** Why were deep shrapnel wounds almost guaranteed to become severely infected within hours? [KT5.3]
- [ ] **43.** What was biologically unique about the mud in Flanders and France that made it so deadly when dragged into a wound? [KT5.3]
- [ ] **44.** What is the defining, fatal characteristic of the 'Gas Gangrene' infection? [KT5.3]
- [ ] **45.** Why did traditional 19th-century surgical antiseptics (like carbolic acid spray) fail to cure Gas Gangrene on the Western Front? [KT5.3]
- [ ] **46.** How did the British Army successfully reduce the threat of Tetanus infections on the Western Front? [KT5.3]
- [ ] **47.** At the start of the war in 1914, why were head injuries disproportionately high (accounting for roughly 20% of wounds)? [KT5.3]
- [ ] **48.** What crucial piece of protective equipment was introduced in 1915, drastically reducing fatal head injuries by 80%? [KT5.3]
- [ ] **49.** What terrifying new weapon did the German army introduce at the Second Battle of Ypres in April 1915? [KT5.3]
- [ ] **50.** What was the physical effect of a Chlorine Gas attack on a soldier? [KT5.3]

Turn page for Vault Part 2 (Questions 51 to 100)

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Complete Knowledge Retrieval Vault · Questions 51 to 100 (Chain of Evacuation & Medical Advances)

VAULT PART 2

- [ ] **51.** Before official gas masks were widely issued in July 1915, how did soldiers desperately attempt to survive gas attacks? [KT5.3]
- [ ] **52.** Which poisonous gas, introduced later in 1915, was significantly faster-acting than chlorine and capable of killing an exposed person within two days? [KT5.3]
- [ ] **53.** What made Mustard Gas (introduced in 1917) such a horrific and distinct weapon? [KT5.3]
- [ ] **54.** Despite the immense psychological terror it caused, why did poison gas actually account for under 5% of all British deaths (around 6,000 soldiers)? [KT5.3]
- [ ] **55.** Which debilitating environmental illness was spread by the millions of body lice living in the seams of the soldiers' unwashed uniforms? [KT5.3]
- [ ] **56.** How did the British Army attempt to medically prevent the spread of Trench Foot in the flooded trenches? [KT5.3]
- [ ] **57.** What was 'Shellshock'? [KT5.3]
- [ ] **58.** Which specialist medical facility in Edinburgh was set up specifically to treat British officers suffering from severe Shellshock? [KT5.3]
- [ ] **59.** What exact acronym did the British Army officially use in medical records to categorise cases of Shellshock? [KT5.3]
- [ ] **60.** Why did the brutal environment of the trenches force the Royal Army Medical Corps (RAMC) to dedicate huge resources to 'preventative measures'? [KT5.3]
- [ ] **61.** What was the fundamental purpose of the 'Chain of Evacuation' on the Western Front? [KT5.4]
- [ ] **62.** Who had the dangerous task of initially rescuing wounded men directly from No Man's Land under enemy fire? [KT5.4]
- [ ] **63.** Where was the Regimental Aid Post (RAP) physically located? [KT5.4]
- [ ] **64.** What was the medical capability of the Regimental Medical Officer stationed at the RAP? [KT5.4]
- [ ] **65.** Who staffed the Advanced Dressing Stations (ADS) and Main Dressing Stations (MDS)? [KT5.4]
- [ ] **66.** How did the British military respond to the violent, injury-worsening journeys caused by horse-drawn ambulance wagons? [KT5.4]
- [ ] **67.** Why did the RAMC still have to rely heavily on horse-drawn wagons (sometimes needing six horses) even after acquiring motor ambulances? [KT5.4]
- [ ] **68.** How were massive numbers of stabilized casualties (up to 800 at a time) transported from the CCS to the Base Hospitals? [KT5.4]
- [ ] **69.** Where were the Casualty Clearing Stations (CCS) strategically located? [KT5.4]
- [ ] **70.** What vital medical sorting process took place at the Casualty Clearing Station (CCS)? [KT5.4]
- [ ] **71.** Because deadly gas gangrene developed so rapidly, how did the role of the Casualty Clearing Station (CCS) critically change during the war? [KT5.4]
- [ ] **72.** Where were Base Hospitals located on the Western Front? [KT5.4]
- [ ] **73.** As the war progressed and the CCS took over emergency amputations, what did Base Hospitals begin to specialize in? [KT5.4]
- [ ] **74.** To cope with the unprecedented scale of casualties, how drastically did the RAMC (Royal Army Medical Corps) expand between 1914 and 1918? [KT5.4]
- [ ] **75.** Who were the QAIMNS, whose numbers grew from 300 to 10,000 during the war? [KT5.4]
- [ ] **76.** What does the acronym FANY stand for? [KT5.4]
- [ ] **77.** What was the most famous and physically demanding frontline role undertaken by the female volunteers of the FANY? [KT5.4]
- [ ] **78.** Aside from driving ambulances, what other vital morale and hygiene services did the FANY provide? [KT5.4]
- [ ] **79.** What was incredibly unique about 'Thompson's Cave', built by British and Commonwealth troops in 1916? [KT5.4]
- [ ] **80.** Why was the underground hospital at Arras highly significant for treating casualties? [KT5.4]
- [ ] **81.** Why was traditional 19th-century aseptic surgery effectively impossible to maintain on the Western Front? [KT5.5]
- [ ] **82.** Faced with rampant gas gangrene, surgeons began using a technique called 'Debridement' (Wound Excision). What did this involve? [KT5.5]
- [ ] **83.** What was the 'Carrel-Dakin method' used to treat deep infections? [KT5.5]
- [ ] **84.** What was the main logistical difficulty with the highly effective Carrel-Dakin solution? [KT5.5]
- [ ] **85.** If antiseptics, debridement, and the Carrel-Dakin method all failed to stop the rapid spread of gas gangrene, what was a surgeon's only remaining option to save the patient's life? [KT5.5]
- [ ] **86.** Before 1915, what was the estimated mortality (death) rate for a soldier who suffered a fractured femur (thigh bone) from a gunshot or shrapnel wound? [KT5.5]
- [ ] **87.** Why did a fractured femur cause such a high death rate during early transport? [KT5.5]
- [ ] **88.** Who originally designed the Thomas Splint before the outbreak of the First World War? [KT5.5]
- [ ] **89.** How did the Thomas Splint (introduced to the Western Front by Robert Jones in December 1915) miraculously increase the survival rate for thigh fractures from 20% to 82%? [KT5.5]
- [ ] **90.** What diagnostic technology was absolutely essential for accurately locating deeply embedded shrapnel so surgeons could remove it before infection set in? [KT5.5]
- [ ] **91.** How was X-Ray technology adapted to deal with the vast number of casualties closer to the frontline? [KT5.5]
- [ ] **92.** What was a major technical limitation of the early mobile X-ray units used on the Western Front? [KT5.5]
- [ ] **93.** Early blood transfusions (such as those pioneered by Dr Lawrence Bruce Robertson) were heavily limited because blood could not be stored. How did this force doctors to perform transfusions? [KT5.5]
- [ ] **94.** In 1915, what chemical did Richard Lewisohn discover could be added to blood to successfully stop it from clotting? [KT5.5]
- [ ] **95.** Following Lewisohn's discovery, Richard Weil discovered that blood mixed with sodium citrate could be refrigerated for up to two days. What monumentally significant breakthrough did Francis Rous and James Turner make in 1916? [KT5.5]
- [ ] **96.** Which American doctor utilized these chemical discoveries to establish the world's first 'blood depot' (blood bank) before the Battle of Cambrai in 1917? [KT5.5]
- [ ] **97.** How did Oswald Hope Robertson successfully store the 22 units of pre-donated universal Type O blood for his pioneering blood bank at Cambrai? [KT5.5]
- [ ] **98.** Which pioneering surgeon developed early plastic surgery techniques to rebuild the severe, disfiguring facial injuries caused by shrapnel? [KT5.5]
- [ ] **99.** Why did the brutal environment and massive casualty numbers of the Western Front force rapid medical advancements? [KT5.5]
- [ ] **100.** Which of the following represents a major diagnostic limitation of the mobile X-ray units? [KT5.5]

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Official Mark Scheme & Knowledge Vault Answers · Answers 1 to 50 (Self & Peer Marking Bank)

MARK SCHEME 1

💡 **Quick-Marking Bank:** Cover this bank with your hand or a sheet of paper to test yourself against Page 2, or use for rapid peer marking. Answers 1–50

1. Surgeons moved from 'antiseptic' surgery (killing germs already in the wound) to 'aseptic' surgery (preventing germs from entering the operating theatre in the first place). [✓][X]

2. The steam autoclave (invented in 1881) used to sterilise surgical instruments. [✓][X]

3. 1895 [✓][X]

4. The radiation doses were 1,500 times stronger than today, causing severe burns and hair loss. [✓][X]

5. The glass tubes were incredibly fragile, the machines were too heavy to move easily, and taking a single image could take up to 90 minutes. [✓][X]

6. He discovered three blood groups (A, B, and O), meaning doctors could finally match donor and patient blood types to prevent fatal rejection. [✓][X]

7. Blood coagulated (clotted) as soon as it left the body, meaning it could not be stored in a bank. Transfusions had to be done with the donor and patient connected directly by a tube. [✓][X]

8. The flat agricultural land had its drainage systems destroyed by constant artillery, turning the battlefield into a horrific, waterlogged, muddy swamp. [✓][X]

9. A piece of land that juts out into enemy territory, making it highly dangerous because it is vulnerable to attack from three sides. [✓][X]

10. The Battle of the Somme [✓][X]

11. An extensive underground hospital built into the chalk tunnel network underneath the city, sheltering the wounded from artillery fire. [✓][X]

12. The first ever blood bank, using stored blood to treat soldiers suffering from severe shock. [✓][X]

13. The soldiers' wounds were already deeply infected with anaerobic bacteria from the heavily manured, muddy soil the moment they were injured in the trenches. [✓][X]

14. The deep mud, craters, and destroyed roads made motorized transport impossible near the front, forcing stretcher-bearers to manually carry the wounded for miles. [✓][X]

15. They were completely unprepared for the unique environmental illnesses, the massive casualty numbers, and the horrific shrapnel wounds caused by high-explosive artillery. [✓][X]

16. A static, defensive conflict where millions of men lived for months in unhygienic dugouts, exposed to weather, rats, and human waste while trying to wear down the enemy. [✓][X]

17. The deep mud was so thick that wagons had to be pulled by six horses instead of two, and the severe shaking often worsened the patients' injuries. [✓][X]

18. A complex network of zigzag dugouts consisting of a front line, support trench, reserve trench, and communication trenches linking them. [✓][X]

19. 1907 [✓][X]

20. High-tech medicine (aseptic surgery, x-rays, blood typing) had emerged, but key limitations in mobility and blood storage meant it was not yet adapted for a muddy, static battlefield. [✓][X]

21. Standing for days in cold, waterlogged mud without changing socks. [✓][X]

22. The feet would swell, blister, turn gangrenous, and often require surgical amputation to save the soldier's life. [✓][X]

23. By ordering soldiers to change their socks twice a day and systematically rub their feet with whale oil. [✓][X]

24. A flu-like illness characterized by severe joint pain, shivering, and high fevers. [✓][X]

25. The millions of body lice living in the seams of the soldiers' unwashed uniforms. [✓][X]

26. They set up massive delousing stations and bathhouses to disinfect clothing with hot steam. [✓][X]

27. A vulnerable 'bulge' in the Allied line surrounded by the enemy on three sides, allowing the Germans to fire down from higher ground. [✓][X]

28. They held onto the salient to stop the Germans advancing to the sea, successfully retaining control of the vital English Channel ports. [✓][X]

29. Chlorine gas [✓][X]

30. Offensive mining to tunnel under and blow up the German positions from below. [✓][X]

31. To relieve extreme military pressure on the French army fighting at Verdun. [✓][X]

32. The British suffered nearly 60,000 casualties in a single day, completely overwhelming the medical services. [✓][X]

33. They dug a 2.5-mile network of tunnels sheltering 25,000 troops, which included a fully functioning underground hospital. [✓][X]

34. It was where troops would retreat to if the frontline came under heavy attack or was overrun. [✓][X]

35. To prevent enemy fire or the blast wave from an exploding artillery shell from travelling straight down the line. [✓][X]

36. It made it incredibly difficult and physically exhausting for stretcher-bearers to safely manoeuvre seriously wounded men around the tight corners while under fire. [✓][X]

37. The heavy motor ambulances frequently got permanently stuck in the deep, waterlogged mud. [✓][X]

38. The delay caused men to die from blood loss or shock, and allowed deep infections like gas gangrene to take hold before they could reach a Casualty Clearing Station. [✓][X]

39. A stomach infection causing severe diarrhea and dehydration, spread by drinking contaminated water from shell holes. [✓][X]

40. That the relentless psychological trauma, noise, and sheer terror of the artillery war caused severe mental breakdowns, resulting in tiredness, nightmares, and uncontrollable shaking. [✓][X]

41. High-explosive artillery shells and shrapnel. [✓][X]

42. The jagged shrapnel tore through flesh, physically dragging pieces of muddy, bacteria-soaked uniform deep into the wound. [✓][X]

43. It was agricultural farmland that had been heavily fertilized with animal manure, packing the soil with the anaerobic bacteria that caused tetanus and gas gangrene. [✓][X]

44. It was a rapid infection that produced a foul-smelling gas inside dying muscle tissue, turning the flesh black and potentially killing a man within a single day. [✓][X]

45. Antiseptics only cleaned the surface of a wound; they could not reach the lethal bacteria that had been driven deep into the muscle by the force of an explosion. [✓][X]

46. By administering routine anti-tetanus injections to wounded soldiers from late 1914 onwards. [✓][X]

47. Soldiers were fighting in trenches; their bodies were protected by earth, but their heads were exposed while wearing only soft cloth caps. [✓][X]

48. The steel Brodie helmet [✓][X]

49. Chlorine Gas [✓][X]

50. It destroyed the respiratory system, causing victims to slowly suffocate as their lungs filled with fluid. [✓][X]

THE BRITISH SECTOR OF THE WESTERN FRONT, 1914–1918

Official Mark Scheme & Knowledge Vault Answers · Answers 51 to 100 (Self & Peer Marking Bank)

MARK SCHEME 2

💡 **Quick-Marking Bank:** Use for rapid recall checking against Vault Part 2 (Page 3). Log total correct recall items on Page 1. Answers 51–100

51. They urinated on cotton pads or handkerchiefs and pressed them to their faces to chemically neutralize the chlorine. [✓][X]

52. Phosgene Gas [✓][X]

53. It was completely odourless and worked slowly, causing severe internal and external blisters that could burn skin straight through uniforms. [✓][X]

54. The rapid development and distribution of highly effective gas masks successfully protected the vast majority of troops. [✓][X]

55. Trench Fever [✓][X]

56. By ordering soldiers to change their socks twice a day and systematically rub their feet with whale oil. [✓][X]

57. A poorly understood psychological trauma (PTSD) caused by the constant bombardment of war, resulting in nightmares, loss of speech, and uncontrollable shaking. [✓][X]

58. Craiglockhart Hospital [✓][X]

59. NYD.N (Not Yet Diagnosed, Nervous) [✓][X]

60. Because the harsh conditions caused so many massive casualties from illnesses (like trench foot and fever) that they had to actively prevent disease just to keep the army fit enough to fight. [✓][X]

61. To provide a highly organised, multi-stage system of medical triage to rescue, transport, and treat massive numbers of casualties without overwhelming frontline units. [✓][X]

62. Teams of four to six stretcher-bearers, who carried the casualties by hand through the mud. [✓][X]

63. Within 200m of the frontline, often in a communication trench or dugout. [✓][X]

64. They could only provide immediate first aid (like bandages and tourniquets) to get the walking wounded back to the fight or prepare serious cases for transport; they could not perform surgery. [✓][X]

65. Units of the Field Ambulance, consisting of medical officers, orderlies, and eventually nurses. [✓][X]

66. The Times newspaper launched a public appeal in 1914, raising money to purchase 512 new motor ambulances. [✓][X]

67. Because heavy motor vehicles frequently got bogged down and entirely stuck in the deep, waterlogged mud of the battlefield. [✓][X]

68. On specially designed ambulance trains and canal barges. [✓][X]

69. 7 to 12 miles back from the frontline, safely out of artillery range but positioned near railway lines for rapid transport. [✓][X]

70. Triage, where patients were sorted into three groups: the walking wounded, those needing hospital treatment, and those with no chance of recovery. [✓][X]

71. It took over the role of performing critical, life-saving surgery (like amputations) from the Base Hospitals to stop the infection before transport. [✓][X]

72. Near the French and Belgian coastal ports, ready to ship patients back to 'Blighty' (Britain). [✓][X]

73. The longer-term recovery of patients and the development of specialist wards for complex issues like head wounds or gas poisoning. [✓][X]

74. From 9,000 men to 113,000 men. [✓][X]

75. Queen Alexandra's Imperial Military Nursing Service, a corps of highly trained professional military nurses. [✓][X]

76. First Aid Nursing Yeomanry. [✓][X]

77. They drove the motor ambulances, navigating terrible roads and deep mud under active shellfire to transport the wounded. [✓][X]

78. They transported supplies, set up cinemas, and operated mobile bath units that could bathe up to 40 men an hour. [✓][X]

79. It was a massive, fully functioning underground hospital built into existing chalk tunnels beneath the town of Arras. [✓][X]

80. Because it functioned as a fully equipped facility incredibly close to the frontline (with 700 stretcher beds, electricity, and water), while remaining completely safe from enemy artillery. [✓][X]

81. Because the chaotic, muddy casualty stations were impossible to keep perfectly sterile, and traditional carbolic acid did not kill gas gangrene bacteria effectively. [✓][X]

82. Aggressively and rapidly cutting away all dead, damaged, and infected tissue from around a wound to physically remove the gangrene before stitching it up. [✓][X]

83. A system where tubes pumped a chemical sterilized salt solution directly into deep wounds to continually wash out and kill the infection-causing bacteria. [✓][X]

84. The solution only stayed fresh for six hours and had to be constantly made as it was needed. [✓][X]

85. Immediate amputation of the infected limb (resulting in over 240,000 lost limbs by 1918). [✓][X]

86. 80% [✓][X]

87. Because traditional splints did not keep the leg straight, meaning the jagged broken bones ground together, severing major arteries and causing fatal internal bleeding and shock. [✓][X]

88. Hugh Owen Thomas [✓][X]

89. It rigidly pulled the leg lengthways (traction), completely stopping the broken bones from grinding on each other and preventing fatal blood loss and shock during transport. [✓][X]

90. X-Rays [✓][X]

91. Six mobile X-ray units (vans) were developed to operate in the British sector, allowing them to be driven directly to Casualty Clearing Stations. [✓][X]

92. The delicate glass tubes overheated very quickly, meaning the machines could only be used for an hour before needing to cool down. [✓][X]

93. Transfusions had to be done 'arm-to-arm', directly transferring blood from a live donor to the patient before it clotted. [✓][X]

94. Sodium Citrate [✓][X]

95. They added citrate glucose to the blood, which allowed it to be safely stored for up to four weeks. [✓][X]

96. Oswald Hope Robertson [✓][X]

97. He stored the blood in glass bottles packed with ice and sawdust. [✓][X]

98. Harold Gillies [✓][X]

99. Because traditional peacetime surgical methods completely failed to treat the complex, infected shrapnel wounds, forcing doctors to innovate radical new techniques out of sheer desperation. [✓][X]

100. They could not detect pieces of dirty clothing that had been driven into wounds by shrapnel, which often caused lethal infections. [✓][X]

Round 1 Exam Practice begins on Page 6

Question 1 (a): Feature Question (2 Marks)
F-D Formula · ~3 Mins

Describe one feature of the work of Casualty Clearing Stations (CCS) on the Western Front.

🚀 **Level 1 Launchpad:** Starter: *"One feature of Casualty Clearing Stations was that..."*
Facts to Include: Located 7–12 miles back near railways; triage (walking wounded, urgent surgery, dying); performed critical amputations and chest surgeries before base evacuation.

One feature of Casualty Clearing Stations was that...

Specifically, (add one precise historical detail or statistic)...

Question 1 (b): Feature Question (2 Marks)
F-D Formula · ~3 Mins

Describe one feature of the use of the Thomas splint on the Western Front.

🚀 **Level 1 Launchpad:** Starter: *"One feature of the Thomas splint was that..."*
Facts to Include: Introduced in 1915 by Robert Jones; rigid frame pulled leg straight to stop bone ends grinding; dramatically dropped compound fracture mortality from 80% to 20%.

One feature of the Thomas splint was that...

Specifically, (add one precise historical detail or statistic)...

Question 2 (b): Follow-Up Investigation Grid (4 Marks)
4-Step Enquiry · ~6 Mins

Study Source B (Edith Smith letter: *"The pipes freeze entirely, meaning we must strictly ration the water provided to the wounded."*). How could you follow up Source B to find out more about the severe conditions faced by medical staff? Complete the table below.

Detail in Source B that I would follow up:	*"The pipes freeze entirely, meaning we must strictly ration the water provided to the wounded."*
Question I would ask:	*What specific impact did water shortages have on surgical sterilization and infection rates in winter 1917?*
What type of source I would use:	*Base Hospital Daily Medical Superintendent Logbooks and Sanitary Inspection Reports (1917–18).*
How this source would help me answer my question:	*It would provide official daily records of sanitation breakdowns and hospital infection rates during freezing months.*

Section A · Turn page for Question 2(a) 8-Mark Source Utility

Question 2 (a): Source Utility (8 Marks)

C-O-P Formula · ~12 Mins

Study Sources A and B. How useful are Sources A and B for an historian studying the immense challenges of treating casualties on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context.

Source A (Chaplain Arthur Davies, Somme CCS, July 1916):

"The casualties have started to arrive... The volume of patients is overwhelming; every bed is occupied, and men are fortunate just to find a spot on the bare ground. Many will pass away before surgery is possible. We have admitted 1,500 casualties, and the stream has not stopped."

Source B (Edith Smith, VAD Nurse, Base Hospital, Dec 1917):

"Our morning duties begin at 3.30 am! The freezing temperatures are unbearable. Icicles hang thick over ward windows. Even kettles, rubber hot water bottles, and sponges have frozen solid. The pipes freeze entirely, meaning we must strictly ration water provided to the wounded."

🔍 **PROVENANCE CLUES (Author, Audience, Motive):** Who wrote each source (eyewitness chaplain at CCS vs frontline VAD nurse at base hospital)? What was their motive (private diary recording sheer volume vs private letter home detailing physical conditions)? How does their perspective make the sources useful despite subjective emotional focus?

PARAGRAPH 1: UTILITY OF SOURCE A (CONTENT + PROVENANCE + CONTEXT)

Source A is useful for investigating casualty challenges because the content reveals...

From my own knowledge, this accurately reflects the Somme offensive because...

Furthermore, the provenance makes it useful because as a private chaplain's diary...

PARAGRAPH 2: UTILITY OF SOURCE B & COMPARATIVE JUDGEMENT (CONTENT + PROVENANCE + CONTEXT)

Source B is also useful because it highlights the severe environmental difficulties...

Specifically, my own knowledge confirms that Base Hospitals in winter 1917...

The provenance enhances its utility because as a private letter from a VAD nurse...

Overall, both sources are mutually useful because together they demonstrate...

⚡ **Level 3 (Grade 9) Utility Glue:** Weigh up typicality vs limitation — note that Source A shows peak crisis conditions on 1 July 1916 (57,000 casualties in one day), while Source B reflects seasonal winter extremes at the coast.

Questions 1(a) & 1(b): Features (2 + 2 = 4 Marks)

Independent Toolkit

🧰 Fact Vault (AO1)

- **Mobile X-rays:** 6 mobile units in British sector; located at CCS; couldn't detect clothing fragments; tubes overheated.
- **Trench Fever:** Caused by body lice; flu-like pain in legs; affected ~15% of men; led to delousing stations and bathhouses.

⚠️ **Trap:** Do not confuse Trench Fever (lice) with Trench Foot (waterlogged mud/cold).

1(a) Describe one feature of mobile X-ray units near the frontline:

One feature of mobile X-ray units was...

1(b) Describe one feature of trench fever:

One feature of trench fever was...

Question 2(b): Follow-Up Practice (4 Marks)

Historical Records Directory

Study this extract from an RAMC officer at Ypres: *"The chlorine gas casualties stumbled in clutching their throats. We washed their blinded eyes with bicarbonate of soda solutions, but many died from asphyxiation."*

Detail in Source to follow up:	*"We washed their blinded eyes with bicarbonate of soda solutions..."*
Question I would ask:	*How effective were early chemical antidotes compared to later respirator gas masks?*
What type of source I would use:	*RAMC Field Ambulance Casualty Admission Books & War Office Gas Defence Bulletins (1915).*
How this source would help me:	*It would record official medical survival rates of gas victims before and after the issue of Small Box Respirators.*

Section A · Turn page for Question 2(a) Source Utility Masterclass

Question 2(a): Source Utility Independent Execution (8 Marks)

Wartime Trauma & Gas Warfare

How useful are Source C (RAMC Logbook, 1915) and Source D (Official Training Manual on Trench Foot, 1916) for an enquiry into the prevention of trench illness on the Western Front?

⏱ Provenance Evaluation Matrix

Official Reports: High factual accuracy regarding procedures; may minimize failure to maintain morale.

Doctor/Nurse Accounts: Direct clinical insight into actual suffering; localized to one specific sector.

Causal & Utility Connectives

- *"The content is highly useful because it reveals..."*
- *"The provenance strengthens its value because..."*
- *"However, the source is limited in scope because..."*

🎯 **Level 3 Standard:** Explicitly evaluate how the author's professional role and purpose shape what was included or omitted.

PARAGRAPH 1: UTILITY OF SOURCE C

Source C is useful for an enquiry into prevention because...

Its provenance as an RAMC logbook means...

PARAGRAPH 2: UTILITY OF SOURCE D & SYNTHESIS

Source D is useful in a different way because it shows official policy on...

However, an official manual may not reflect actual frontline reality because...

Comparing both sources, an historian learns that...

Section A · Round 3 Exam Simulation begins on Page 10

SECTION A: THE HISTORIC ENVIRONMENT

Round 3: The Planning Engine Room · Deconstruction Matrix (Plan here, write on Page 11)

ROUND 3: ENGINE ROOM

1. Question 1 Features Selector:

Exam Prompt	Feature 1 (Core Concept)	Supporting Specific Fact / Detail (AO1)
Q1(a) Underground Hospital at Arras	Built into chalk quarries; fully equipped	700 beds, running water, electricity, operating theatre, mortuary.
Q1(b) Blood Transfusion at Cambrai	First large-scale indirect blood bank (1917)	Oswald Robertson used sodium citrate/dextrose; treated 20 severely wounded men.

2. Question 2(a) Source Utility Planning Grid:

Source	Content Knowledge (What it reveals)	Provenance Analysis (NOP & Evaluation)
Source A (Medical)	Specific wound types (shrapnel, compound fracture, gangrene)	Doctor/nurse eyewitness account; high clinical validity but localized.
Source B (Soldier)	Stretcher bearer delays in mud, shell craters, shell shock	Infantry diary; emotional impact; subject to wartime censorship.

3. Question 2(b) Follow-Up Pre-Flight Selector:

Detail to Quote	Targeted Question	Authentic Source Type	How it Answers
Exact phrase about delays in stretcher evacuation	What was the average time taken from RAP to CCS in the Ypres salient?	RAMC Field Ambulance Section War Diaries (WO 95)	Provides recorded times of stretcher bearer transfers across muddy terrain.

📋 **PRE-FLIGHT AUDIT CHECKLIST:** [ ] Did I state TWO separate features for Q1? [ ] Did I analyze BOTH content and provenance for Sources A & B? [ ] Is my Q2(b) source a specific historical record (not 'the internet')?

SECTION A: THE HISTORIC ENVIRONMENT

TIMED EXAM PITCH

AO1 Knowledge: Precise names, locations (Ypres, Somme, Cambrai), equipment, and dates.

AO3 Sources: Balanced analysis of utility using content, contextual knowledge, and provenance.

Question 1(a) (2 Marks):*Describe one feature of the underground hospital at Arras:* ______________________

Question 1(b) (2 Marks):*Describe one feature of blood transfusion techniques on the Western Front:* _________

Question 2(a) (8 Marks):*Source A is useful because* ____

Source B is also useful because ____

Question 2(b) (4 Marks): Complete the follow-up investigation table below:

Detail in Source B to follow up:	
Question I would ask:	
What type of source I would use:	
How this source would help me:	

Section A · Turn page for 100% Spec Coverage Bank

⚠️ EXAMINER TRAP DOORS & 100% SPECIFICATION BANK

Section A: The British Sector of the Western Front · Pearson Edexcel GCSE History

100% Spec Guarantee

🚫 TOP 3 FATAL EXAMINER TRAPS TO AVOID FOR SECTION A

• The "Generic Source" Trap
Never write "I would check a diary" or "the internet" in Q2(b). You must specify an authentic record type, like RAMC War Diaries or Hospital Admission Registers.

• The "Treachery of Bias" Trap
Never reject a source simply because "it is biased so it's useless". Sources written with strong emotion or wartime censorship are highly useful for showing attitudes!

• The Single Feature Trap
Remember Q1 is split into two questions: 1(a) and 1(b). Do not combine them into one paragraph. Each requires one identified feature + one specific factual detail.

📚 SPECIFICATION PRACTICE BANK: 100% CURRICULUM COVERAGE

Every remaining Western Front specification bullet point tested below

1. Features (2m): Describe one feature of the role of the FANY (First Aid Nursing Yeomanry) in ambulance transport.

2. Features (2m): Describe one feature of the effects of mustard gas on soldiers.

3. Source Utility (8m): How useful are official trench casualty statistics compared to private letters for studying the effectiveness of Brodie helmets?

4. Features (2m): Describe one feature of the Carrel-Dakin method for treating gas gangrene.

5. Follow-Up (4m): Detail: *"Ambulance barges along the Somme-Yser canal were fitted with spring beds."* Formulate a question and name a primary source.

⚡ **QUICK-FIRE FOLLOW-UP SOURCE DRILL:** When asked to research: (1) *Wound infections* → RAMC Medical Officer Daily Logs; (2) *Frontline delousing routines* → Battalion War Diaries; (3) *Blood transfusions* → CCS Surgical Register.

PEARSON EDEXCEL GCSE (9–1) HISTORY · PAPER 1 (1HI0/11)

Section B 24-Page Thematic Master

Medicine in Britain, c1250–present

Thematic Study across 750 Years · Medieval, Renaissance, 18th/19th Century & Modern Eras

Pupil Name: ______________________________ **Class / Set:** __________ **Target Grade:** [ ]

📊 SECTION B PROGRESS & ASSESSMENT TRACKER (36+4 MARKS TOTAL)

Marked by Teacher or Peer Verified

Assessment Component & Stem Focus	Format Style	Max Marks	Pupil Score
🧠 **Complete 4-Era Knowledge Vault (All 280 Recall Questions)**	**Checklist**	**/280**	_____
Round 1 (Stepped): Q3 Similarity / Difference (Medieval vs Renaissance)	Stepped Ladder	/4	_____
Round 1 (Stepped): Q4 Explain Why (Surgical Progress c1700–c1900)	Stepped Ladder	/12	_____
Round 1 (Stepped): Q5/Q6 Statement Essay (Germ Theory vs Public Health)	Stepped Ladder	/16+4	_____
Round 2 (Dual-Track): Q3 Similarity / Difference (Great Plague vs Black Death)	Dual Track	/4	_____
Round 2 (Dual-Track): Q4 Explain Why (Penicillin Mass Production c1938–45)	Dual Track	/12	_____
Round 2 (Dual-Track): Q5/Q6 Statement Essay (Four Humours c1250–c1700)	Dual Track	/16+4	_____
Round 3 (Simulation): Section B Full Exam Hall Simulation (Q3, Q4, Q5/6)	Exam Pitch (55m)	/36+4	_____
TOTAL COMBINED EXAM PRACTICE MARKS:		**/104**	_____ **/ 104**

Teacher / Peer Marker: ______________________________ **Date:** ____________ **Overall Grade: [9 8 7 6 5 4]**

⏱ SECTION B EDEXCEL EXAM FAST FACTS & STEM BLUEPRINTS (36+4 MARKS · ~55 MINS) OPTION 11 PAPER 1 SECTION B

- **Q3 Similarity / Difference (4 Marks · ~6 mins):** *P-F-C Formula.* State core **Point of Comparison** in sentence 1 → support with **specific evidence from Period 1** → support with **specific evidence from Period 2** → explain significance.
- **Q4 Explain Why (12 Marks · ~18 mins):** *3 P-E-E Paragraphs.* **Point** clearly linking to the question → **Evidence** deploying precise facts/names/dates → **Explanation** tracing the direct causal link. (Must use both stimulus points + own knowledge).
- **Q5/Q6 Either/Or Statement Essay (16 Marks + 4 SPaG · ~30 mins):** *Balanced Evaluation.* Point 1 (Agree with statement) → Point 2 (Counter-factor/Disagree) → Point 3 (Alternative factor) → **Criteria-Led Conclusion** explaining *why* one factor was more significant.

UNIT 1: MEDICINE IN MEDIEVAL BRITAIN, C1250–C1500

Knowledge Retrieval Vault · Part 1: Questions 1 to 40 (Causes of Illness & Religious Beliefs)

MEDIEVAL VAULT 1

- [ ] **1.** Who was the Ancient Greek physician credited with creating the Theory of the Four Humours? [KT1.1]
- [ ] **2.** In the Theory of the Four Humours, which humour was associated with the season of Autumn, the element of Earth, and the qualities of cold and dry? [KT1.1]
- [ ] **3.** Which humour was linked to the temperament 'sanguine' and had the qualities of being hot and wet? [KT1.1]
- [ ] **4.** How did the 2nd-century Roman physician Galen develop Hippocrates' original ideas on the humours? [KT1.1]
- [ ] **5.** Why did the medieval Christian Church so strongly support and promote Galen's medical ideas? [KT1.1]
- [ ] **6.** What was the name of the mid-13th-century medical textbook collection used by students in European universities? [KT1.1]
- [ ] **7.** According to the medieval Church's teachings, why did God send diseases like leprosy to humans? [KT1.1]
- [ ] **8.** During the medieval period, what did physicians use to determine the best time for treatment based on the positions of the planets? [KT1.1]
- [ ] **9.** In medieval England, an unusual planetary alignment of Mars, Jupiter, and Saturn in the year 1345 was widely blamed for causing which disaster? [KT1.1]
- [ ] **10.** Which theory suggested that breathing in 'bad air' filled with harmful fumes from swamps, corpses, or rotting matter corrupted the humours? [KT1.1]
- [ ] **11.** Why did the Church strictly discourage and prevent medical students and physicians from performing human dissections? [KT1.1]
- [ ] **12.** In the Theory of the Four Humours, which element and qualities were associated with Phlegm? [KT1.1]
- [ ] **13.** What was the primary purpose of a medieval physician examining a patient's urine? [KT1.1]
- [ ] **14.** Which three characteristics of a urine sample did medieval physicians check on a urine chart wheel? [KT1.1]
- [ ] **15.** Which ancient Greek philosopher's work on logic and observation was studied by medieval medical students alongside the works of Galen? [KT1.1]
- [ ] **16.** In the context of the Theory of Opposites, how would a medieval physician treat a patient who had a cold, wet fever (excess phlegm)? [KT1.1]
- [ ] **17.** How did the Church's total control over education and book production affect medical progress in the Middle Ages? [KT1.1]
- [ ] **18.** What was the 'Vademecum' carried by medieval physicians? [KT1.1]
- [ ] **19.** In the Theory of the Four Humours, the liver was associated with which humour? [KT1.1]
- [ ] **20.** How was leprosy viewed by the medieval Church and population? [KT1.1]
- [ ] **21.** What was the most common medical method used to balance an excess of the blood humour in medieval England? [KT1.2]
- [ ] **22.** Which of the following was a standard method of bloodletting where a vein was opened, usually in the arm, using a sharp tool called a fleam? [KT1.2]
- [ ] **23.** In what medical scenario would a medieval physician choose leeching (using leeches) over venesection? [KT1.2]
- [ ] **24.** During the cupping method of bloodletting, what was the purpose of heating the glass or metal cup before placing it on the patient's skin? [KT1.2]
- [ ] **25.** What was the medical purpose of 'purging' a patient in medieval medicine? [KT1.2]
- [ ] **26.** What was a 'clyster' in medieval medicine? [KT1.2]
- [ ] **27.** What was the 'Regimen Sanitatis'? [KT1.2]
- [ ] **28.** Which of the following was a highly popular, complex medieval remedy containing up to 70 herbs, spices, and ingredients like opium and snake flesh, used as a universal antidote? [KT1.2]
- [ ] **29.** How did medieval people attempt to prevent miasma from entering their bodies or homes? [KT1.2]
- [ ] **30.** Why did medieval physicians often recommend bathing in warm water as a treatment? [KT1.2]
- [ ] **31.** If a patient was suffering from a hot and dry illness, such as a fever, how would a physician apply the Theory of Opposites to treat them? [KT1.2]
- [ ] **32.** What religious act was commonly undertaken by sick medieval people to show repentance for their sins and beg God for a miraculous physical cure? [KT1.2]
- [ ] **33.** What did the medieval Church teach was the spiritual value of enduring physical pain and illness? [KT1.2]
- [ ] **34.** Which herbal preparation involved crushing medicinal plants and applying them directly to the skin to draw out poisons or ease pain? [KT1.2]
- [ ] **35.** Why did public baths ('stewes') in England begin to close down toward the end of the medieval period? [KT1.2]
- [ ] **36.** In the context of humoural medicine, why did physicians pay close attention to what patients ate and drank? [KT1.2]
- [ ] **37.** What was the primary method of purging a patient if the physician believed the excess humour was located in the upper stomach? [KT1.2]
- [ ] **38.** What spiritual practice did extreme religious groups perform publicly during crises to appease God's anger and prevent disease? [KT1.2]
- [ ] **39.** Which of these was a common laxative plant used in medieval England to purge the bowels? [KT1.2]
- [ ] **40.** What was the purpose of keeping a piece of amber or gold near a patient suffering from certain illnesses? [KT1.2]

Turn page for Medieval Vault Part 2

- [ ] **41.** Which medical practitioner in medieval England attended university for up to ten years, was licensed to diagnose illness, but rarely carried out physical treatment himself? [KT1.2]
- [ ] **42.** Which group of medieval medical practitioners mixed herbal remedies, sold spices, and were often viewed as a cheaper alternative to expensive physicians? [KT1.2]
- [ ] **43.** What training route did medieval barber-surgeons take to learn their craft, such as performing amputations or pulling teeth? [KT1.2]
- [ ] **44.** Why did the medieval Catholic Church strictly ban monks and priests from performing surgeries that involved shedding blood? [KT1.2]
- [ ] **45.** In a typical medieval hospital run by the Church, what was the primary focus of the monastic staff? [KT1.2]
- [ ] **46.** Which of the following groups of patients would be systematically turned away and rejected from entering a medieval charity hospital? [KT1.2]
- [ ] **47.** Why did many medieval hospitals have their chapel positioned at the very center of the ward, visible from all patient beds? [KT1.2]
- [ ] **48.** In medieval England, who was responsible for providing the vast majority of daily medical care and herbal treatments to the sick? [KT1.2]
- [ ] **49.** Which of the following was a manual surgical task that a medieval barber-surgeon would regularly perform? [KT1.2]
- [ ] **50.** Why were university-trained physicians so rarely employed or seen in medieval England? [KT1.2]
- [ ] **51.** What was the main diagnostic tool a medieval physician used at a patient's bedside before consulting their books? [KT1.2]
- [ ] **52.** How did apothecaries typically acquire their knowledge of herbs, spices, and drug preparation? [KT1.2]
- [ ] **53.** Why did medieval physicians rarely touch or physically examine their patients during a consultation? [KT1.2]
- [ ] **54.** Which of these ingredients was commonly grown in a medieval home garden and used by women to treat basic coughs and chest ailments? [KT1.2]
- [ ] **55.** How did the Guild of Barber-Surgeons regulate the standard of surgery in medieval towns? [KT1.2]
- [ ] **56.** What was a major limitation of a medieval apothecary's remedies? [KT1.2]
- [ ] **57.** Approximately how many charity hospitals existed in medieval England by the year 1400? [KT1.2]
- [ ] **58.** In what way did medieval charity hospitals contribute to public health? [KT1.2]
- [ ] **59.** What did medieval surgeons use to try and dull the pain of an amputation, despite the extreme risk of killing the patient? [KT1.2]
- [ ] **60.** Why did the medieval medical community believe that academic physicians were of a higher social and intellectual standing than barber-surgeons? [KT1.2]
- [ ] **61.** In which year did the Black Death first arrive on the shores of England? [KT1.3]
- [ ] **62.** What is the actual scientific cause of the Black Death, and how was it primarily spread during the Middle Ages? [KT1.3]
- [ ] **63.** Which astrological event in the year 1345 was widely blamed by medieval scholars for causing the Black Death? [KT1.3]
- [ ] **64.** What was the name of the hard, painful, pus-filled swellings that appeared in the armpits and groin of Black Death victims? [KT1.3]
- [ ] **65.** Which lung-based variant of the plague was spread directly through coughing and sneezing, was 100% fatal, and killed victims much faster? [KT1.3]
- [ ] **66.** Why did the religious groups known as 'flagellants' whip themselves publicly in the streets of London? [KT1.3]
- [ ] **67.** What natural disaster did medieval people believe had cracked open the earth, releasing the poisonous fumes (miasma) that caused the plague? [KT1.3]
- [ ] **68.** Which French physician to the Pope famously advised people to 'Go quickly, go far, and return slowly' to survive the Black Death? [KT1.3]
- [ ] **69.** Under the local quarantine rules introduced during the Black Death, how long were travelers new to an area forced to stay away from others? [KT1.3]
- [ ] **70.** Why did local authorities in some medieval towns deliberately stop sweeping and cleaning the streets during the Black Death? [KT1.3]
- [ ] **71.** Why were medieval local governments largely unsuccessful in fully enforcing quarantine laws during the 1348 epidemic? [KT1.3]
- [ ] **72.** Which traditional treatment did medieval physicians try on Black Death patients that actually made them die more quickly? [KT1.3]
- [ ] **73.** Which treatment occasionally allowed a Black Death patient to survive if it was done successfully by a barber-surgeon? [KT1.3]
- [ ] **74.** Approximately what percentage of the English population is estimated to have died during the Black Death epidemic of 1348-49? [KT1.3]
- [ ] **75.** In which year did King Edward III write his famous letter to the Mayor of London complaining about the filthy, death-carrying air of the city? [KT1.3]
- [ ] **76.** Which popular, spice-based medieval medicine containing up to 70 ingredients (including opium) was prescribed as an antidote during the Black Death? [KT1.3]
- [ ] **77.** Which of the following was a common way people tried to prevent 'bad air' (miasma) from entering their bodies during the Black Death? [KT1.3]
- [ ] **78.** Why is it historically incorrect to write about 'searchers', 'red crosses on doors', or 'killing 40,000 dogs' in a GCSE answer about the 1348 Black Death? [KT1.3]
- [ ] **79.** Why did the plague return to England periodically every 10 to 20 years after the first 1348 outbreak? [KT1.3]
- [ ] **80.** Where do historians believe the Black Death first broke out before traveling along trade routes to Sicily, Europe, and finally England? [KT1.3]

Turn page for Renaissance Knowledge Vault

UNIT 2: THE MEDICAL RENAISSANCE IN BRITAIN, C1500–C1700

Knowledge Retrieval Vault · Part 1: Questions 1 to 30 (Ideas, Printing Press & Sydenham)

RENAISSANCE VAULT 1

- ☐ **1.** What does the French word 'Renaissance' translate to, and what did it represent in medical history?
[KT2.1]
- ☐ **2.** How did the rise of Protestantism during the Reformation affect the Church's influence over scientific and medical learning in the Renaissance?
[KT2.1]
- ☐ **3.** How did Johannes Gutenberg's printing press (developed in around 1440) directly support progress in medical knowledge during the Renaissance?
[KT2.1]
- ☐ **4.** Why was the 17th-century English physician Thomas Sydenham known as the 'English Hippocrates'?
[KT2.1]
- ☐ **5.** How did Thomas Sydenham's view of disease fundamentally challenge the traditional Theory of the Four Humours?
[KT2.1]
- ☐ **6.** Which two diseases did Thomas Sydenham successfully distinguish and classify as separate illnesses in his clinical observation work?
[KT2.1]
- ☐ **7.** What was the primary objective of the Royal Society, which was founded in London in the year 1660?
[KT2.1]
- ☐ **8.** What was the official Latin motto of the Royal Society, and what did it encourage scientists to do?
[KT2.1]
- ☐ **9.** What was the name of the Royal Society's scientific journal, first published in 1665, which allowed scientists to share and peer-review research globally?
[KT2.1]
- ☐ **10.** In 1683, Antony van Leeuwenhoek used an improved microscope to observe 'animalcules' (bacteria) for the first time. What was the major limitation of this breakthrough in the Renaissance?
[KT2.1]
- ☐ **11.** Why did Renaissance physicians continue to use the Theory of the Four Humours when diagnosing patients, despite many doctors beginning to doubt its scientific accuracy by the late 17th century?
[KT2.1]
- ☐ **12.** How did beliefs about 'miasma' change during the Medical Renaissance?
[KT2.1]
- ☐ **13.** Andreas Vesalius's anatomical dissections proved that Galen had made over 300 mistakes. Why did this breakthrough have almost no immediate impact on everyday medical treatments?
[KT2.1]
- ☐ **14.** Which of the following was a revolutionary treatment for smallpox developed by Thomas Sydenham that directly challenged traditional medieval practices?
[KT2.1]
- ☐ **15.** What was the 'Royal Touch' and what did its popularity during the Renaissance demonstrate about society's beliefs?
[KT2.1]
- ☐ **16.** What intellectual movement in the Renaissance period encouraged people to think for themselves, ask questions, and use observation to find the truth?
[KT2.1]
- ☐ **17.** How did the training of physicians in universities begin to improve during the Renaissance?
[KT2.1]
- ☐ **18.** In what way did the printing press improve scientific accuracy in medical research compared to manual copying by medieval monks?
[KT2.1]
- ☐ **19.** What popular New World herbal remedy, imported from Peru, did Thomas Sydenham popularise to treat malaria (ague)?
[KT2.1]
- ☐ **20.** Which of the following statements best summarises the progress made in ideas about the causes of disease between 1500 and 1700?
[KT2.1]
- ☐ **21.** What is the primary reason why Andreas Vesalius's major Renaissance breakthroughs in anatomy did not immediately improve everyday medical treatments for patients?
[KT2.2]
- ☐ **22.** Which of the following treatments represents a major CONTINUITY in everyday medical practice between the Medieval period and the Medical Renaissance (c1500-c1700)?
[KT2.2]
- ☐ **23.** What was the new Renaissance medical theory of 'transference', and how was it practically applied by patients?
[KT2.2]
- ☐ **24.** How did exploration of the 'New World' (the Americas) change herbal medicine in Renaissance England?
[KT2.2]
- ☐ **25.** What was the medical term 'iatrochemistry' (or alchemy), which became increasingly popular during the Renaissance?
[KT2.2]
- ☐ **26.** Why did the chemical element antimony become a controversial treatment during the Renaissance?
[KT2.2]
- ☐ **27.** Which major political event in 1536 had a devastating immediate impact on hospital care and charity beds across England?
[KT2.2]
- ☐ **28.** Following the Dissolution of the Monasteries in 1536, what major change occurred in how surviving hospitals were funded and run?
[KT2.2]
- ☐ **29.** How did the primary focus of hospitals shift during the Renaissance compared to their medieval predecessors?
[KT2.2]
- ☐ **30.** How did the London city council and local charity manage to keep St Bartholomew's Hospital open after 1536?
[KT2.2]

Turn page for Renaissance Vault Part 2

UNIT 2: THE MEDICAL RENAISSANCE IN BRITAIN, C1500–C1700

Knowledge Retrieval Vault · Part 2: Questions 31 to 60 (Vesalius, Harvey & The Great Plague 1665)

RENAISSANCE VAULT 2

- [ ] **31.** What were 'Pest Houses' and why were they introduced during the Renaissance period? [KT2.2]
- [ ] **32.** If a Renaissance patient suffered from a disease that produced a red skin rash, such as smallpox, what traditional herbal-associative treatment would they most likely receive? [KT2.2]
- [ ] **33.** What was a major change in the medical staff employed by municipal hospitals like St Bartholomew's during the late Renaissance? [KT2.2]
- [ ] **34.** Why did everyday people in Renaissance England continue to rely on traditional family treatments in the home rather than municipal hospitals or professional physicians? [KT2.2]
- [ ] **35.** Which of the following is a classic example of iatrochemistry/alchemy used as an everyday treatment in the late 1600s, showing the dangerous nature of some Renaissance medical experiments? [KT2.2]
- [ ] **36.** How did the role of professional guilds, such as the Company of Barber-Surgeons, affect medical training during the Renaissance? [KT2.2]
- [ ] **37.** How did the treatment of plague victims in the 1665 Great Plague demonstrate continuity with the 1348 Black Death? [KT2.2]
- [ ] **38.** Why did people in the 1500s and 1600s begin to take fewer warm baths, representing a decline in hygiene compared to the medieval period? [KT2.2]
- [ ] **39.** What did King Charles II's final medical treatment in 1685 (such as opening veins for bleeding and purging his bowels) demonstrate about the state of royal medicine at the end of the Renaissance? [KT2.2]
- [ ] **40.** Which statement best describes the overall pace of change in medical TREATMENTS during the Medical Renaissance (c1500-c1700)? [KT2.2]
- [ ] **41.** In his landmark 1628 publication, An Anatomical Account of the Motion of the Heart and Blood, what did William Harvey empirically prove about the human body? [KT2.3]
- [ ] **42.** What experiment did William Harvey use on a patient's arm to prove that blood flows in one direction through the veins? [KT2.3]
- [ ] **43.** Which ancient authority's medical teachings did William Harvey's work on the circulatory system directly contradict and disprove? [KT2.3]
- [ ] **44.** Why did William Harvey's discovery of blood circulation have almost no immediate impact on everyday medical treatments in the short term? [KT2.3]
- [ ] **45.** How did the English medical establishment initially react to William Harvey's publication in 1628? [KT2.3]
- [ ] **46.** How long did it take for William Harvey's discoveries about blood circulation to be officially adopted into the medical curriculum at English universities? [KT2.3]
- [ ] **47.** What vital scientific limitation prevented William Harvey from fully explaining how blood traveled from the arteries back into the veins? [KT2.3]
- [ ] **48.** In the long term, why was William Harvey's breakthrough in physiology so significant for modern medicine? [KT2.3]
- [ ] **49.** How did the mortality rate of the 1665 Great Plague in London compare to the 1348 Black Death in England? [KT2.3]
- [ ] **50.** What was the most significant difference in the government and municipal response to the 1665 Great Plague compared to the 1348 Black Death? [KT2.3]
- [ ] **51.** If a London household had a member diagnosed with the Great Plague in 1665, what quarantine measure was legally enforced by watchmen? [KT2.3]
- [ ] **52.** What was the specific, dangerous duty of the municipal officials known as 'searchers' during the Great Plague of 1665? [KT2.3]
- [ ] **53.** Why did London authorities order the mass slaughter of roughly 40,000 dogs and cats in 1665, and what was the unintended consequence? [KT2.3]
- [ ] **54.** Which of the following was a preventative measure enforced by municipal authorities in London in 1665 to limit the spread of the plague? [KT2.3]
- [ ] **55.** How did municipal authorities organize the burial of plague victims in 1665 to minimize the risk of miasma? [KT2.3]
- [ ] **56.** Why did both medieval mayors in 1348 and Renaissance mayors in 1665 order giant barrels of tar and bonfires to be burned in the streets? [KT2.3]
- [ ] **57.** What was the believed physical purpose of the bird-shaped beak mask worn by Renaissance 'plague doctors'? [KT2.3]
- [ ] **58.** Which bizarre, superstitious treatment recorded during the 1665 Great Plague demonstrates the complete lack of improvement in medical cures since 1348? [KT2.3]
- [ ] **59.** Why were schoolboys at Eton College severely punished or beaten if they refused to smoke tobacco during the Great Plague of 1665? [KT2.3]
- [ ] **60.** What does the continued use of amulets, carrying pomanders, and humoural bleeding during the Great Plague of 1665 demonstrate about Renaissance medicine? [KT2.3]

Turn page for 18th & 19th Century Knowledge Vault

UNIT 3: MEDICINE IN 18TH & 19TH-CENTURY BRITAIN, C1700–C1900

Knowledge Retrieval Vault · Part 1: Questions 1 to 30 (Germ Theory, Pasteur, Koch & Surgery)

18TH/19TH VAULT 1

- [ ] **1.** Before Louis Pasteur's Germ Theory was accepted, what was the popular scientific theory that claimed microscopic organisms were the result rather than the cause of decay? [KT3.1]
- [ ] **2.** How did the theory of Spontaneous Generation differ from the traditional theory of Miasma? [KT3.1]
- [ ] **3.** In which year did Louis Pasteur publish his landmark Germ Theory, and what was he originally investigating that led to this discovery? [KT3.1]
- [ ] **4.** Which famous piece of laboratory equipment did Louis Pasteur use to prove that microbes in the air caused decay, rather than the air itself creating them? [KT3.1]
- [ ] **5.** How did the publication of Germ Theory in 1861 immediately affect everyday medical treatments for the general public in Britain? [KT3.1]
- [ ] **6.** Why did many traditional British doctors and the British government initially reject Pasteur's Germ Theory in the 1860s? [KT3.1]
- [ ] **7.** How did the German scientist Robert Koch build upon Louis Pasteur's Germ Theory of disease? [KT3.1]
- [ ] **8.** Which breakthrough laboratory methodology did Robert Koch develop that allowed scientists to grow and study 'pure' cultures of specific bacteria without them getting mixed up? [KT3.1]
- [ ] **9.** Why did Robert Koch begin using synthetic aniline dyes in his bacteriological research? [KT3.1]
- [ ] **10.** Which deadly disease-causing microbe did Robert Koch successfully identify and isolate in the year 1882, which was the biggest killer in Victorian Britain? [KT3.1]
- [ ] **11.** Which of the following is a common chronological error regarding Robert Koch's use of microscopes? [KT3.1]
- [ ] **12.** How did Robert Koch's work fundamentally change the way British medical scientists studied diseases? [KT3.1]
- [ ] **13.** Which British surgeon was among the first to read Louis Pasteur's 1861 Germ Theory and apply its principles to solve the problem of infection in surgery? [KT3.1]
- [ ] **14.** In 1870, which physicist supported Pasteur's Germ Theory by theorizing that disease-carrying germs were spread through microscopic dust particles in the air? [KT3.1]
- [ ] **15.** In what way did the work of Pasteur and Koch eventually lead to massive progress in the prevention of disease by the late 19th century? [KT3.1]
- [ ] **16.** How did the intense scientific and political rivalry between France (Pasteur) and Germany (Koch) affect medical progress? [KT3.1]
- [ ] **17.** According to the Theory of Spontaneous Generation, where did microbes come from? [KT3.1]
- [ ] **18.** How did Louis Pasteur's swan-neck flask experiments specifically disprove Spontaneous Generation? [KT3.1]
- [ ] **19.** Why is it historically inaccurate to state that Edward Jenner developed his 1796 smallpox vaccine as a direct result of Pasteur's Germ Theory? [KT3.1]
- [ ] **20.** In 1883, Robert Koch traveled to Egypt and Calcutta to identify the specific microbe causing which deadly water-borne epidemic? [KT3.1]
- [ ] **21.** What was the critical difference between the 1848 Public Health Act and the 1875 Public Health Act? [KT3.2]
- [ ] **22.** What did Florence Nightingale believe was the primary cause of disease spreading in hospitals when she reformed them? [KT3.2]
- [ ] **23.** Which of the following correctly pairs a 19th-century surgical breakthrough with its specific medical function? [KT3.2]
- [ ] **24.** Which feature of Florence Nightingale's 'pavilion-style' hospital design was specifically meant to combat the threat of miasma? [KT3.2]
- [ ] **25.** In which year did James Simpson discover the anaesthetic effects of chloroform, and what was its immediate impact on surgery? [KT3.2]
- [ ] **26.** How did Joseph Lister develop his carbolic acid antiseptic spray in 1865? [KT3.2]
- [ ] **27.** Why did the discovery of chloroform as an anaesthetic initially lead to the 'Black Period' of surgery (1840s-1870s)? [KT3.2]
- [ ] **28.** What did the 19th-century British government's 'laissez-faire' attitude refer to, and how did it affect public health? [KT3.2]
- [ ] **29.** Where did Florence Nightingale establish the world's first professional training school for nurses in 1860, and what was its focus? [KT3.2]
- [ ] **30.** Why did the 1848 Public Health Act fail to significantly improve living conditions across most of Britain? [KT3.2]

Turn page for 18th & 19th Century Vault Part 2

- [ ] **31.** By 1890, the medical community shifted from 'antiseptic' surgery to 'aseptic' surgery. What is the difference between these two approaches? [KT3.2]
- [ ] **32.** What did the tragic death of 14-year-old Hannah Greener in 1848 during a minor toenail operation demonstrate about early anaesthetics? [KT3.2]
- [ ] **33.** Which of the following was a major reason for initial opposition to James Simpson's use of chloroform in childbirth and surgery? [KT3.2]
- [ ] **34.** Which historical event in 1853 finally broke down public and medical resistance to the use of chloroform in Britain? [KT3.2]
- [ ] **35.** Which of the following duties was compulsory for local city authorities to provide under the 1875 Public Health Act? [KT3.2]
- [ ] **36.** How did Edwin Chadwick's 1842 Report on the Sanitary Conditions of the Labouring Population help pave the way for public health reform? [KT3.2]
- [ ] **37.** What was the significance of the 'Great Stink' of 1858 in London? [KT3.2]
- [ ] **38.** During her work at the military hospital in Scutari during the Crimean War, Florence Nightingale's hygienic reforms successfully dropped the mortality rate by what margin? [KT3.2]
- [ ] **39.** What did Joseph Lister find was the long-term impact of his carbolic spray, even though the spray itself fell out of favor by 1890? [KT3.2]
- [ ] **40.** Why did many surgeons initially refuse to use Joseph Lister's carbolic acid antiseptics in the 1860s? [KT3.2]
- [ ] **41.** In which year did Edward Jenner develop the smallpox vaccine, and what traditional preventative method did it replace? [KT3.3]
- [ ] **42.** What was the process of smallpox inoculation used in 18th-century Britain, and what was its major clinical risk? [KT3.3]
- [ ] **43.** On what specific group of people did Edward Jenner make his initial observations that led to the discovery of the smallpox vaccine? [KT3.3]
- [ ] **44.** What empirical experiment did Jenner perform in 1796 to prove the safety and effectiveness of his smallpox vaccine? [KT3.3]
- [ ] **45.** What was a massive scientific limitation of Edward Jenner's 1796 vaccine breakthrough? [KT3.3]
- [ ] **46.** Why was Jenner's vaccine a 'one-off' breakthrough in the late 18th and early 19th centuries? [KT3.3]
- [ ] **47.** Which of the following is a common student misconception regarding Edward Jenner's cowpox discovery? [KT3.3]
- [ ] **48.** How did the British government eventually support Edward Jenner's smallpox vaccine in the 19th century? [KT3.3]
- [ ] **49.** Despite government backing, why was there initial public and professional opposition to Jenner's smallpox vaccine in Britain? [KT3.3]
- [ ] **50.** Which deadly disease did Dr. John Snow investigate in London in 1854, and what was its popular believed cause before his work? [KT3.3]
- [ ] **51.** What was John Snow's theory about how cholera was transmitted, which he published in his 1849 pamphlet? [KT3.3]
- [ ] **52.** How did John Snow gather empirical evidence to prove his water-borne cholera theory during the 1854 Broad Street epidemic? [KT3.3]
- [ ] **53.** What famous 'control group' case studies did John Snow use to prove that the 1854 Soho cholera outbreak was linked to the water pump? [KT3.3]
- [ ] **54.** What immediate, physical action did John Snow take in Soho that stopped the local 1854 cholera outbreak? [KT3.3]
- [ ] **55.** How did the General Board of Health and the wider medical establishment initially react to John Snow's 1854 findings? [KT3.3]
- [ ] **56.** What major environmental event in 1858, combined with Snow's research, finally forced the British government to build a massive sewer system for London? [KT3.3]
- [ ] **57.** Who was the engineer appointed by the government to construct London's massive new sewer system (completed in 1875) following John Snow's work and the Great Stink? [KT3.3]
- [ ] **58.** Why is it a common examiner error for students to state that John Snow's 1854 discovery led to an *immediate* national change in public health laws? [KT3.3]
- [ ] **59.** Which subsequent scientific developments finally provided the undeniable biological proof that John Snow's water-borne cholera theory was correct? [KT3.3]
- [ ] **60.** Comparing Jenner's and Snow's breakthroughs, what makes them similar in terms of 19th-century medical progress? [KT3.3]

Turn page for Modern Medicine Knowledge Vault

UNIT 4: MEDICINE IN MODERN BRITAIN, C1900–PRESENT

Knowledge Retrieval Vault · Part 1: Questions 1 to 40 (Genetics, DNA, Diagnosis & Magic Bullets)

MODERN VAULT 1

- ☐ **1.** How did the 20th-century understanding of what causes illness fundamentally differ from the 19th-century focus on Germ Theory? [KT4.1]
- ☐ **2.** Which 19th-century scientist first demonstrated that physical characteristics could be passed down between generations of living things? [KT4.1]
- ☐ **3.** What crucial scientific technique did Rosalind Franklin and Maurice Wilkins use at King's College London to help identify the structure of DNA? [KT4.1]
- ☐ **4.** In which year did James Watson and Francis Crick successfully work out and model the double-helix structure of DNA? [KT4.1]
- ☐ **5.** What major international scientific project was launched in 1990 to identify and map every single gene in human DNA? [KT4.1]
- ☐ **6.** What is a major modern limitation of the genetic breakthroughs achieved by mapping the human genome? [KT4.1]
- ☐ **7.** How do modern doctors use DNA discoveries to prevent or catch illnesses early through genetic screening? [KT4.1]
- ☐ **8.** According to modern scientific research, what specific medical condition can be triggered by eating a diet that is too high in sugar? [KT4.1]
- ☐ **9.** What chronic modern disease is directly linked to eating a diet too high in fat and experiencing obesity? [KT4.1]
- ☐ **10.** Apart from lung cancer, what other debilitating respiratory disease has modern research linked directly to tobacco smoking? [KT4.1]
- ☐ **11.** Which chronic, organ-based illnesses has modern science linked directly to the excessive drinking of alcohol? [KT4.1]
- ☐ **12.** Which major piece of technology, invented in the 1930s, was 1,000 times more powerful than standard light microscopes, allowing scientists to see viruses for the first time? [KT4.1]
- ☐ **13.** How did diagnosis in the modern era fundamentally change compared to previous historical periods? [KT4.1]
- ☐ **14.** When were blood tests first introduced into modern medicine, and what was their diagnostic significance? [KT4.1]
- ☐ **15.** What is the key diagnostic advantage of a modern CT scan compared to a traditional X-ray? [KT4.1]
- ☐ **16.** When were ultrasound scans first developed, and how do they help doctors diagnose internal conditions like gallstones or kidney stones? [KT4.1]
- ☐ **17.** What are ECGs (electrocardiograms), which were developed in the 1900s, used for in modern diagnosis? [KT4.1]
- ☐ **18.** What is an endoscope (such as a bronchoscope used in lung investigations), and how does it assist in modern diagnosis? [KT4.1]
- ☐ **19.** What is a biopsy, and what role does it play in the modern diagnosis of chronic illnesses like cancer? [KT4.1]
- ☐ **20.** When was blood sugar monitoring introduced, and how did it change the daily management of diabetes? [KT4.1]
- ☐ **21.** In 1909, Paul Ehrlich and his research team developed Salvarsan 606 to treat syphilis. Why was Salvarsan 606 clinically known as the first 'magic bullet'? [KT4.2]
- ☐ **22.** During his research, how did Paul Ehrlich discover Salvarsan 606? [KT4.2]
- ☐ **23.** In 1932, Gerhard Domagk discovered Prontosil, the second 'magic bullet'. What type of infection did Prontosil cure? [KT4.2]
- ☐ **24.** How did 20th-century 'magic bullets' fundamentally differ in their clinical application from 19th-century antiseptics like Joseph Lister's carbolic acid? [KT4.2]
- ☐ **25.** What is the biological difference between early chemical 'magic bullets' like Salvarsan 606 and the first true antibiotic, penicillin? [KT4.2]
- ☐ **26.** After Gerhard Domagk published his findings on Prontosil, what active ingredient did French scientists isolate, leading to the creation of cheap sulphonamide drugs? [KT4.2]
- ☐ **27.** Before the creation of the NHS, the Liberal Government introduced the 1911 National Insurance Act. What was a major limitation of this Act's medical coverage? [KT4.2]
- ☐ **28.** Which landmark 1942 government report recommended that the British state should provide comprehensive social security and healthcare 'from the cradle to the grave'? [KT4.2]
- ☐ **29.** In which year did the National Health Service (NHS) begin operating, and who was the government minister responsible for its launch? [KT4.2]
- ☐ **30.** What was the revolutionary founding principle of the National Health Service (NHS) in 1948? [KT4.2]
- ☐ **31.** When the NHS was launched in 1948, its services were organized into a 'tripartite' (three-part) system. What were these three parts? [KT4.2]
- ☐ **32.** Why did the British Medical Association (BMA)-representing the country's doctors-initially strongly oppose the creation of the NHS? [KT4.2]
- ☐ **33.** How did Aneurin Bevan famously win over the doctors who opposed the NHS? [KT4.2]
- ☐ **34.** How did the launch of the NHS in 1948 transform access to high-tech medical treatments for the working class? [KT4.2]
- ☐ **35.** Following the introduction of sulphonamide magic bullets in the 1930s and the NHS in 1948, what dramatic trend occurred in maternal mortality (mothers dying during or after childbirth)? [KT4.2]
- ☐ **36.** As part of its modern focus on preventing disease, what type of campaigns does the government run through the NHS? [KT4.2]
- ☐ **37.** Why does the 1935 illustration 'Medical moments in time' in the textbook emphasize that the pace of medical change in the first half of the 20th century was steady rather than rapid? [KT4.2]
- ☐ **38.** Which of the following is a successful example of a modern, government-funded NHS preventative immunization campaign that wiped out a major paralyzing disease in Britain? [KT4.2]
- ☐ **39.** Which of the following represents a modern, high-tech surgical treatment developed in the late 20th century that is provided free of charge under the NHS? [KT4.2]
- ☐ **40.** What is a major modern challenge facing the long-term effectiveness of antibiotic treatments like penicillin? [KT4.2]

Turn page for Modern Vault Part 2

UNIT 4: MEDICINE IN MODERN BRITAIN, C1900–PRESENT

Knowledge Retrieval Vault · Part 2: Questions 41 to 80 (Penicillin, NHS 1948 & Lung Cancer)

MODERN VAULT 2

- [ ] **41.** In which year did Alexander Fleming make his accidental observation of Penicillium mould on a staphylococci culture plate, and where did this take place? [KT4.3]
- [ ] **42.** What did Alexander Fleming actually observe in his forgotten petri dish in 1928 that led to the discovery of penicillin? [KT4.3]
- [ ] **43.** Why did Alexander Fleming abandon his research into penicillin as a systemic medical treatment by 1931? [KT4.3]
- [ ] **44.** Which of the following is a widespread historical 'myth' regarding the development of penicillin that examiners warn students to avoid in their essays? [KT4.3]
- [ ] **45.** How did Alexander Fleming's early laboratory experiments in 1928-1929 mislead him about penicillin's potential as an internal medicine? [KT4.3]
- [ ] **46.** Which two scientists revived Alexander Fleming's neglected 1929 paper on penicillin in 1938, and where were they based? [KT4.3]
- [ ] **47.** How did Ernst Chain contribute to the team's early breakthroughs in purifying penicillin? [KT4.3]
- [ ] **48.** What was the result of Florey and Chain's landmark scientific experiment on mice in the year 1940? [KT4.3]
- [ ] **49.** Why did Florey and Chain's Oxford research team have to use household objects like milk churns, bed pans, and even a bath tub in their laboratory? [KT4.3]
- [ ] **50.** Who was the first human patient to be treated with Florey and Chain's purified penicillin in 1941, and how did he contract his infection? [KT4.3]
- [ ] **51.** What tragic clinical outcome occurred during the first human trial of penicillin in 1941, and why? [KT4.3]
- [ ] **52.** When Florey and Chain first approached British pharmaceutical companies in 1939-1940 to mass-produce penicillin, why were they turned down? [KT4.3]
- [ ] **53.** To which country did Howard Florey travel in July 1941 to seek the industrial help and investment needed to mass-produce penicillin? [KT4.3]
- [ ] **54.** What major historical event in December 1941 served as a catalyst, prompting the US government to heavily fund the mass production of penicillin? [KT4.3]
- [ ] **55.** Which industrial manufacturing method allowed US pharmaceutical companies to scale up penicillin production from small laboratory bottles to millions of doses? [KT4.3]
- [ ] **56.** How many doses of penicillin did Allied medical services have available by the D-Day landings in June 1944, demonstrating the success of wartime mass-production? [KT4.3]
- [ ] **57.** How did Howard Florey's attitude toward patenting penicillin accelerate its worldwide availability and keep manufacturing costs down? [KT4.3]
- [ ] **58.** Which female scientist at Oxford University identified the exact chemical structure of penicillin in 1945, allowing chemists to create semi-synthetic variations? [KT4.3]
- [ ] **59.** In which year did Fleming, Florey, and Chain jointly receive the Nobel Prize in Medicine to recognize their collective roles in the penicillin breakthrough? [KT4.3]
- [ ] **60.** Looking at the penicillin story through Edexcel GCSE 'factors', which combination of factors was most crucial in turning a minor 1928 laboratory discovery into a global medical revolution by 1944? [KT4.3]
- [ ] **61.** In which year did the British Medical Research Council publish the landmark study by Doll and Hill that conclusively proved the link between smoking and lung cancer? [KT4.4]
- [ ] **62.** Why was the British government initially very slow to take legislative action against smoking after the scientific link to lung cancer was proven in 1950? [KT4.4]
- [ ] **63.** What was a major limitation of using traditional chest X-rays to diagnose lung cancer before advanced scanning technology was developed? [KT4.4]
- [ ] **64.** What key technological advantage does a modern CT (computerised tomography) scan have over a traditional chest X-ray in diagnosing lung cancer? [KT4.4]
- [ ] **65.** How does a high-tech PET (positron emission tomography) scan assist modern doctors in managing a lung cancer patient's treatment? [KT4.4]
- [ ] **66.** What is bronchoscopy (a type of endoscope) and how is it used to diagnose lung cancer? [KT4.4]
- [ ] **67.** What is a biopsy, and why is it a crucial step in the modern diagnosis of lung cancer? [KT4.4]
- [ ] **68.** How does modern chemotherapy treat lung cancer, and what is its main clinical drawback? [KT4.4]
- [ ] **69.** What does modern radiotherapy treatment for lung cancer involve? [KT4.4]
- [ ] **70.** What is immunotherapy, one of the most advanced treatments for lung cancer in the twenty-first century? [KT4.4]
- [ ] **71.** What was the very first major advertising restriction the British government placed on tobacco, introduced in 1965? [KT4.4]
- [ ] **72.** Following gradual extensions of advertising rules, in which year did the government finally implement a complete ban on all tobacco advertising and sports sponsorship in the UK? [KT4.4]
- [ ] **73.** In 2007, what legislative measure did the government take to restrict teenagers' access to tobacco products? [KT4.4]
- [ ] **74.** Under the Health Act of 2006, what major legislative ban came into force on 1 July 2007 to protect the public from passive smoking? [KT4.4]
- [ ] **75.** What retail display regulation was introduced in 2012 to discourage young people from taking up smoking? [KT4.4]
- [ ] **76.** In 2015, how did the government extend the smoking ban to protect children from the dangers of second-hand smoke? [KT4.4]
- [ ] **77.** Which of the following was a 'nudge' tactic introduced in 2015 that standardizes cigarette boxes to make them unappealing to young people? [KT4.4]
- [ ] **78.** Why is there currently no national routine screening program for lung cancer in the UK, unlike other chronic conditions? [KT4.4]
- [ ] **79.** When comparing government action against 19th-century cholera with 20th-century lung cancer, what major similarity do historians identify? [KT4.4]
- [ ] **80.** The government's modern campaigns (like the Sugar Tax and Change4Life) and legislative bans represent a transition away from which historical political philosophy? [KT4.4]

SECTION B: OFFICIAL MARK SCHEME

Answers-Only Bank: Medieval (Answers 1–80) · Quick-Marking Centerfold

MEDIEVAL ANSWERS

💡 **Quick-Marking Bank:** Cover with your hand or exercise book to test your recall against Pages 2–3.

1. Hippocrates [✓][X]	**2.** Black bile [✓][X]	**3.** Blood [✓][X]
4. By creating the Theory of Opposites to treat humoural imbalances with their opposite qualities [✓][X]	**5.** Because his writings argued the body was perfectly designed by a single 'Creator', fitting Christian theology [✓][X]	**6.** The Articella [✓][X]
7. As a punishment for sins or as a test of an individual's religious faith [✓][X]	**8.** An Almanac (astrology chart) [✓][X]	**9.** The arrival of the Black Death [✓][X]
10. The Miasma theory [✓][X]	**11.** Because Church doctrine taught that the physical body needed to be buried whole to go to heaven [✓][X]	**12.** Water; cold and wet [✓][X]
13. To check the internal balance of the patient's four humours [✓][X]	**14.** Color, thickness, and smell [✓][X]	**15.** Aristotle [✓][X]
16. By giving them hot, dry foods or spices like hot peppers [✓][X]	**17.** It hindered progress by preserving and copying only classical texts, preventing challenge to Galen [✓][X]	**18.** A handbook of medical diagrams and charts used during diagnosis [✓][X]
19. Yellow bile (choler) [✓][X]	**20.** As a terrifying skin disease sent as a direct punishment from God for sins [✓][X]	**21.** Bloodletting (phlebotomy) [✓][X]
22. Venesection [✓][X]	**23.** For patients who were too young, elderly, or physically weak to withstand a vein being opened [✓][X]	**24.** To create a vacuum that would pull blood to the surface of the skin [✓][X]
25. To expel excess or corrupt humours from the digestive tract using emetics or laxatives [✓][X]	**26.** An enema used to clear the bowels, often containing honey, water, and herbal mixtures [✓][X]	**27.** A personalized set of rules regarding diet, exercise, sleep, and hygiene written by a physician to help a patient prevent illness [✓][X]
28. Theriaca [✓][X]	**29.** By carrying sweet-smelling herbs, burning incense, or carrying a pomander [✓][X]	**30.** To help sweat out excess humours, open the pores, and dissolve impurities in the body [✓][X]
31. By prescribing cool, wet remedies, such as giving them cucumbers or cold baths [✓][X]	**32.** Going on a pilgrimage to a holy shrine containing saintly relics [✓][X]	**33.** It was a way to cleanse the soul of sin and shorten a person's time in Purgatory [✓][X]
34. A poultice [✓][X]	**35.** Because they became associated with moral corruption, prostitution, and the spread of diseases like leprosy [✓][X]	**36.** Because different foods had specific qualities (hot, cold, wet, dry) that directly influenced and balanced the humours [✓][X]
37. Administering an emetic to induce vomiting [✓][X]	**38.** Flagellation (whipping themselves to show repentance for sin) [✓][X]	**39.** Senna [✓][X]
40. To protect the patient through sympathetic magic or transference, believing the gold would draw out the disease [✓][X]	**41.** A university-trained physician [✓][X]	**42.** Apothecaries [✓][X]
43. Serving a practical apprenticeship under an experienced master in a guild [✓][X]	**44.** Because spilling blood was seen as incompatible with a holy, spiritual office [✓][X]	**45.** Providing basic physical care, warmth, food, and spiritual preparation for the soul [✓][X]
46. Infectious people, pregnant women, and the mentally ill [✓][X]	**47.** To ensure patients could observe and participate in daily holy Mass from their beds, prioritizing spiritual healing [✓][X]	**48.** Women in the home, such as mothers, wives, and local wise women [✓][X]
49. Trepanning the skull or setting fractured limbs [✓][X]	**50.** Because they were extremely expensive and their services were limited to the wealthy and royalty [✓][X]	**51.** An inspection of the patient's pulse and urine sample [✓][X]
52. Through hands-on experience, trade networks, and passing down secrets from father to son [✓][X]	**53.** Because they believed medical knowledge was a purely intellectual, book-based science of logical observation [✓][X]	**54.** Mint and chamomile [✓][X]
55. By monitoring apprenticeships, inspecting workshops, and punishing unlicensed or dangerous practitioners [✓][X]	**56.** They often mixed ingredients based on astrological alignments or supernatural properties rather than proven clinical trial [✓][X]	**57.** Over 1,100 [✓][X]
58. By providing a clean, warm environment and nutritious food to help vulnerable people recover naturally [✓][X]	**59.** Herbal mixtures containing highly toxic hemlock, opium, or mandrake [✓][X]	**60.** Because physicians were university-educated and understood the logical theories of Galen, whereas surgeons were considered manual craftsmen who worked with their hands [✓][X]
61. 1348 [✓][X]	**62.** Yersinia pestis bacteria, carried in the digestive systems of fleas on black rats [✓][X]	**63.** An unusual alignment of the planets Mars, Jupiter, and Saturn [✓][X]
64. Buboes [✓][X]	**65.** Pneumonic plague [✓][X]	**66.** To show God they were sorry for their sins and beg Him to stop the plague [✓][X]
67. A volcanic eruption or earthquake [✓][X]	**68.** Guy de Chauliac [✓][X]	**69.** 40 days [✓][X]
70. Because they believed the foul stench of rotting waste would drive away the plague miasma [✓][X]	**71.** Because they lacked power, meaning the rich moved freely and the Church ran services as normal [✓][X]	**72.** Bleeding and purging [✓][X]
73. Cutting open (lancing) the buboes to drain the pus [✓][X]	**74.** Between 30% and 50% (nearly one-third to one-half) [✓][X]	**75.** 1349 [✓][X]
76. Theriaca [✓][X]	**77.** Carrying sweet-smelling herbs, flowers, or spices [✓][X]	**78.** Because those specific administrative preventions occurred only during the 1665 Great Plague [✓][X]
79. Because unhygienic, crowded, and rat-infested living conditions remained completely unchanged [✓][X]	**80.** The Far East (such as China) [✓][X]	

SECTION B: OFFICIAL MARK SCHEME

Answers-Only Bank: Renaissance (Answers 1–60) · Quick-Marking Centerfold

RENAISSANCE ANSWERS

💡 **Quick-Marking Bank:** Score checked items and verify your factual understanding of Harvey, Vesalius and Sydenham.

1. Rebirth; a period of reborn interest in classical worlds combined with a love of enquiry and a willingness to challenge old ideas [✓][X]

2. It made the Catholic Church much less able to promote and enforce its preferred beliefs about science and nature [✓][X]

3. It allowed new, controversial discoveries to be printed rapidly and shared cheaply, making it harder for the Church to censor or block them [✓][X]

4. Because he urged doctors to stop relying blindly on classical books and instead closely observe their patients' symptoms to diagnose them [✓][X]

5. He argued that diseases were separate, independent 'species' that could be grouped and classified like plants, rather than being unique to each individual's humoural balance [✓][X]

6. Measles and Scarlet Fever [✓][X]

7. To encourage scientific enquiry, promote empirical experimentation, and provide a hub for scientists to share their research [✓][X]

8. 'Nullius in verba' (Take nobody's word for it); it encouraged scientists to do experiments to prove things rather than just trusting classical authorities [✓][X]

9. Philosophical Transactions [✓][X]

10. He did not realise that these 'animalcules' actually caused illnesses, meaning spontaneous generation and miasma remained the dominant beliefs [✓][X]

11. Because patients fully understood the humoural system, and physicians had to respect their clients' expectations to earn a living [✓][X]

12. The belief in miasma remained highly popular and became even more widespread during devastating epidemics like the Great Plague of 1665 [✓][X]

13. Because knowing the correct structure of the bones and organs did not give doctors new cures, so they continued to use traditional bleeding and herbal remedies [✓][X]

14. Prescribing airy bedrooms, light blankets, and cold drinks instead of the traditional 'sweating' method [✓][X]

15. The belief that the monarch could cure skin diseases like scrofula simply by touching the patient, demonstrating how slowly traditional and supernatural beliefs changed [✓][X]

16. Humanism [✓][X]

17. Students began to observe real human dissections alongside classical texts, using highly accurate printed anatomical illustrations [✓][X]

18. It guaranteed that every copy of a medical book contained identical, error-free text and precise anatomical drawings [✓][X]

19. Cinchona bark (quinine) [✓][X]

20. New anatomical knowledge and clinical observations laid vital foundations for the future, but everyday medical beliefs and treatments for the general public changed very little [✓][X]

21. Knowing the correct structure of bones and organs did not give doctors new chemical cures or treatments, so they continued to use traditional bleeding and purging [✓][X]

22. The continued reliance on bloodletting and purging to balance the Four Humours [✓][X]

23. The belief that an illness could be transferred to another object or animal, such as rubbing an onion on warts [✓][X]

24. It introduced new, effective herbal ingredients like cinchona bark (quinine) to treat malaria [✓][X]

25. The search for chemical and mineral-based cures (using metals like antimony) rather than plant-based remedies [✓][X]

26. Because in large doses it promoted sweating and vomiting to clear the body, but in small doses it was used as a daily general tonic [✓][X]

27. Henry VIII's Dissolution of the Monasteries [✓][X]

28. They transitioned from monastic control to being run and funded by secular town councils and charities [✓][X]

29. They shifted from providing general hospitality and spiritual 'care' to actively treating wounds and curable physical illnesses [✓][X]

30. By petitioning Henry VIII, who re-founded the hospital himself under municipal control in 1546 [✓][X]

31. Specialized, secular isolation hospitals opened by town councils specifically to quarantine people suffering from highly contagious diseases like the plague [✓][X]

32. The 'red cure,' which involved drinking red wine, eating red foods, and wearing red clothing [✓][X]

33. They began employing university-trained physicians to actively treat and cure patients' physical conditions [✓][X]

34. Because physicians were extremely expensive, and mothers, wives, or local wise women provided familiar, free, and accessible herbal care [✓][X]

35. Prescribing ingredients derived from metals and minerals, such as mercury to treat syphilis, or using 'cold deadman's skull' in remedies [✓][X]

36. They helped to professionalize surgeons by regulating training through apprenticeships, separating their work from simple hair cutting [✓][X]

37. Both outbreaks were treated with humoural remedies (bleeding and purging), herbal mixtures, and prayers, because the cause (germs) was still unknown [✓][X]

38. Because public bath houses became associated with the rapid spread of the 'Great Pox' (syphilis), making people terrified of catching it [✓][X]

39. That despite monumental scientific discoveries, clinical medicine at the bedside remained heavily based on traditional, medieval humoural treatments [✓][X]

40. Treatments remained highly traditional with immense continuity, as the new scientific discoveries in anatomy and physiology did not yet translate into practical bedside cures [✓][X]

41. That the heart acts as a muscular pump circulating blood continuously around the body in a one-way system [✓][X]

42. He tied a tight bandage (ligature) around the arm and showed blood could only be pushed past the valves in one direction [✓][X]

43. Galen's theory that the liver constantly manufactures new blood [✓][X]

44. Knowing that blood circulated did not cure diseases, so physicians continued to use traditional bleeding and purging [✓][X]

45. Many doctors openly criticized or ignored him because his theory had no practical application at the bedside [✓][X]

46. It took nearly 50 years, with his ideas only regularly appearing in university medical lectures from around 1673 [✓][X]

47. Microscopes of the era were not powerful enough to see the tiny capillaries linking arteries and veins [✓][X]

48. It served as the essential foundation for subsequent surgical developments, blood transfusions, and cardiology [✓][X]

49. The Great Plague killed roughly one-fifth of London's population (about 100,000 people), whereas the 1348 Black Death killed nearly 30-50% of the entire country [✓][X]

50. In 1665, local authorities implemented highly organized, centralized regulations enforced by watchmen and searchers [✓][X]

51. The house was boarded up with the entire family locked inside for 28 days, marked with a red cross on the door [✓][X]

52. To inspect homes, identify plague victims, confirm the cause of death, and monitor the spread [✓][X]

53. They believed the animals spread the plague through their breath, but killing them allowed the rat population (which carried the plague-carrying fleas) to explode [✓][X]

54. Banning large public gatherings, closing theatres, and locking down taverns [✓][X]

55. Bodies were collected in carts only at night and buried in deep, mass graves (plague pits) outside the city walls [✓][X]

56. To purify the air and drive away the disease-carrying 'bad air' (miasma) [✓][X]

57. They believed birds attracted the plague, so the mask would draw the disease away from the patient and into the herb-filled beak [✓][X]

58. Strapping a live, plucked chicken to a swollen plague bubo to draw out the poison [✓][X]

59. Tobacco smoke was believed to act as a powerful chemical shield to ward off the deadly plague miasma [✓][X]

60. That without an understanding of germs, people's fundamental beliefs about what caused the plague had not changed since 1348 [✓][X]

Turn page for 18th/19th Century & Modern Answers

SECTION B: OFFICIAL MARK SCHEME

Answers-Only Bank: 18th & 19th Century (Answers 1–60)

18TH/19TH ANSWERS

1. The theory of Spontaneous Generation [✓][X]

2. Spontaneous Generation argued that rotting or decaying matter naturally produced microbes, while Miasma claimed that foul-smelling air transmitted disease. [✓][X]

3. 1861; investigating why liquids like beetroot alcohol and wine went sour [✓][X]

4. Swan-neck flasks that kept sterilized liquids fresh by trapping airborne dust [✓][X]

5. It had very little immediate impact on everyday treatments, as most doctors resisted the theory and continued to use traditional remedies. [✓][X]

6. Because Pasteur was a university chemist, not a licensed medical doctor, and his work focused primarily on food and drink rather than human disease. [✓][X]

7. He identified the specific microbes that caused individual diseases, transforming Germ Theory into practical bacteriology. [✓][X]

8. Growing bacteria on solid agar jelly derived from seaweed in a flat dish [✓][X]

9. To stain the transparent bacteria so they would stand out and be clearly visible under a microscope [✓][X]

10. The Tuberculosis (TB) microbe [✓][X]

11. Believing that Koch was the first person to invent the microscope, when in fact Antonie van Leeuwenhoek had observed 'animalcules' back in the 17th century. [✓][X]

12. They stopped studying the physical symptoms of patients and began studying the specific microbes causing the disease. [✓][X]

13. Joseph Lister [✓][X]

14. John Tyndall [✓][X]

15. It allowed scientists to identify specific bacteria and subsequently develop targeted vaccines for multiple diseases. [✓][X]

16. It accelerated progress because both governments funded research teams who raced to identify the microbes for cholera, pneumonia, and plague. [✓][X]

17. They were created by decaying or rotting matter. [✓][X]

18. By showing that microbes existed in the air first and caused decay, and that if air was kept out, no decay or germs appeared in the liquid. [✓][X]

19. Because Jenner's work occurred over 60 years before Pasteur published Germ Theory, meaning Jenner did not actually understand why his vaccine worked. [✓][X]

20. Cholera [✓][X]

21. The 1848 Act was 'permissive' (optional), whereas the 1875 Act was 'compulsory,' forcing local councils to provide clean water, sewers, and waste disposal. [✓][X]

22. Miasma (bad air), which is why she emphasized high ceilings, large windows, and excellent ventilation. [✓][X]

23. Chloroform as an anaesthetic to eliminate pain; Carbolic acid as an antiseptic to kill microbes and prevent infection. [✓][X]

24. Wards built with high ceilings, large windows, and separate wings to maximize ventilation and airflow. [✓][X]

25. 1847; it eliminated pain during operations but initially led to a 'Black Period' where infection rates and surgical deaths rose. [✓][X]

26. After reading Louis Pasteur's Germ Theory and realizing he could use chemistry to kill airborne microbes before they infected wounds. [✓][X]

27. Because surgeons attempted deeper, more complex operations on unconscious patients, but because they did not yet understand antiseptics, many patients died of severe infections. [✓][X]

28. A 'leave it alone' belief that it was not the government's responsibility or right to interfere in people's daily lives, which blocked sanitation improvements for years. [✓][X]

29. At St Thomas' Hospital in London; focusing on cleanliness, strict hygiene, and raising nursing to a respected, trained profession. [✓][X]

30. Because it was optional (permissive), meaning most local town councils chose not to set up Boards of Health or spend taxes on sewers. [✓][X]

31. Antiseptic surgery focuses on killing germs already in the wound (using carbolic acid), while aseptic surgery focuses on preventing germs from getting into the operating theatre in the first place (using steam-cleaned instruments, gowns, and rubber gloves). [✓][X]

32. That chloroform was difficult to dose correctly and carried a serious risk of fatal overdose. [✓][X]

33. Many religious leaders argued that pain during childbirth was sent by God and that Simpson was interfering with God's design. [✓][X]

34. Queen Victoria chose to use chloroform during the birth of her eighth child, making the anaesthetic highly fashionable and accepted. [✓][X]

35. Clean piped water, sewage disposal systems, public toilets, and street lighting. [✓][X]

36. It argued that living in filth and disease caused early deaths and cost the government money, suggesting that cleaning up cities would save taxpayer funds. [✓][X]

37. The smell of rotting waste in the Thames was so severe that it disrupted parliament, finally forcing MPs to fund Joseph Bazalgette to build London's massive sewer system. [✓][X]

38. From 42% down to 2%. [✓][X]

39. It permanently changed the attitude of surgeons, who finally understood that preventing post-operative infection was their clinical duty. [✓][X]

40. Because they did not accept Louis Pasteur's Germ Theory and did not believe that invisible, microscopic bacteria actually existed. [✓][X]

41. 1796; replacing the dangerous practice of smallpox inoculation [✓][X]

42. Smearing active pus from a smallpox scab into a cut in the skin, which carried a high risk of causing a fatal smallpox outbreak [✓][X]

43. Dairy maids who had contracted cowpox and subsequently seemed immune to smallpox [✓][X]

44. He infected a young boy, James Phipps, with cowpox, and later exposed him to smallpox to prove he was immune [✓][X]

45. Because Germ Theory was still decades away, Jenner did not know *why* his vaccine worked, meaning he could not replicate his method for other diseases [✓][X]

46. Without an understanding of microbes and antibodies, scientists could not systematically design vaccines for other diseases until Louis Pasteur's Germ Theory was accepted [✓][X]

47. All of the above are common misconceptions that examiners frequently identify [✓][X]

48. By awarding Jenner £30,000 in grants and eventually passing a law in 1853 making the vaccine compulsory [✓][X]

49. Professional inoculators feared losing their businesses, some religious leaders objected to injecting animal matter, and Jenner could not scientifically explain why the vaccine worked [✓][X]

50. Cholera; believed to be spread by miasma (foul-smelling bad air) or spontaneous generation [✓][X]

51. It was water-borne, spread by drinking water contaminated with cholera-ridden human waste [✓][X]

52. He created a detailed spot map of Soho, showing that deaths were tightly clustered around the Broad Street water pump [✓][X]

53. A local workhouse and a brewery near Broad Street where workers did not catch cholera because they had private wells or drank only beer [✓][X]

54. He removed the handle of the Broad Street water pump so the public could no longer drink from it [✓][X]

55. They largely rejected his findings and clung to Miasma theory, as Snow lacked the microscopic biological proof that Germ Theory would later provide [✓][X]

56. 'The Great Stink' of 1858, where hot weather exposed rotting sewage in the Thames, disrupting Parliament [✓][X]

57. Joseph Bazalgette [✓][X]

58. Because the government maintained a laissez-faire attitude and did not pass a compulsory public health act until 1875, over 20 years later, after Germ Theory had validated Snow's theory [✓][X]

59. Louis Pasteur's Germ Theory (1861) and Robert Koch's identification of the specific cholera microbe (1883) [✓][X]

60. Both individuals used empirical observation and scientific methods to achieve massive breakthroughs, yet faced initial resistance because their findings challenged traditional beliefs (like Miasma) before Germ Theory was established [✓][X]

Turn page for Modern Medicine Answers

SECTION B: OFFICIAL MARK SCHEME

Answers-Only Bank: Modern Medicine (Answers 1–80)

MODERN ANSWERS

1. It moved to a multi-causal approach, recognizing that while microbes cause infectious diseases, genetics and lifestyle choices cause chronic illnesses. [✓][X]
2. Gregor Mendel [✓][X]
3. X-ray crystallography to take photographs of human cells [✓][X]
4. 1953 [✓][X]
5. The Human Genome Project [✓][X]
6. There is still no cure or highly effective treatment for the majority of genetic conditions. [✓][X]
7. By checking an individual's genetic code to see if they carry genes that put them at higher risk of developing certain cancers or conditions. [✓][X]
8. Type 2 diabetes [✓][X]
9. Heart disease [✓][X]
10. Emphysema [✓][X]
11. Liver disease, kidney disease, and several cancers [✓][X]
12. The Electron microscope [✓][X]
13. It shifted from a doctor's clinical observation of physical symptoms to precise, high-tech laboratory tests and scans. [✓][X]
14. The 1930s; they allowed doctors to test for a vast number of conditions without invasive surgery. [✓][X]
15. A CT scan is a 3D scan that creates detailed images of the inside of the body, allowing doctors to spot tiny tumors when they are the size of a pea. [✓][X]
16. The 1940s; they use high-frequency sound waves to build up a picture of the inside of the body. [✓][X]
17. They use electrical impulses to track and monitor heart activity. [✓][X]
18. A flexible tube with a camera that allows doctors to see inside the body and collect cell samples without major surgery. [✓][X]
19. Taking a tiny sample of tissue from a lump to test it and see if it is cancerous. [✓][X]
20. The 1960s; it allowed patients to regularly check their own blood sugar at home to manage their condition. [✓][X]
21. Because it was a synthetic chemical compound designed to target and destroy a specific disease-causing microbe inside the body without harming the rest of the patient's cells. [✓][X]
22. By methodically testing over 600 arsenic chemical compounds until finding one (the 606th) that specifically destroyed syphilis bacteria. [✓][X]
23. Puerperal (childbed) fever and blood poisoning caused by streptococcus bacteria. [✓][X]
24. Antiseptics killed microbes on external surfaces or open wound tissue, whereas magic bullets were taken internally to target and destroy specific pathogens inside the body without poisoning the patient. [✓][X]
25. Magic bullets were synthetic chemical compounds created in laboratories, whereas antibiotics like penicillin were created using living microorganisms to kill bacteria. [✓][X]
26. Sulphanilamide, which allowed pharmaceutical companies to mass-produce cheap, effective drugs for pneumonia, meningitis, and scarlet fever. [✓][X]
27. It only provided free GP care to sick workers who paid weekly contributions, completely excluding their wives, children, and those who were unemployed or retired. [✓][X]
28. The Beveridge Report [✓][X]
29. 1948; under Aneurin Bevan [✓][X]
30. That comprehensive healthcare should be free at the point of delivery for everyone in Britain, paid for through national taxation. [✓][X]
31. Primary care (GPs, dentists, opticians), hospital services (managed by regional boards), and local authority health services (such as ambulances and health visitors). [✓][X]
32. They feared they would lose their independent status, lose their private patient income, and become state-controlled civil servants. [✓][X]
33. By 'stuffing their mouths with gold'-allowing hospital consultants to continue treating lucrative private patients alongside their NHS work. [✓][X]
34. It made advanced diagnostic tests and specialized surgical treatments free for everyone, removing the fear of bankruptcy from getting ill. [✓][X]
35. It fell rapidly, dropping from high rates in the early 20th century to less than 1% by the late 20th century. [✓][X]
36. Public education and lifestyle campaigns encouraging healthy eating, physical exercise, and reducing tobacco or alcohol consumption. [✓][X]
37. Because while breakthroughs like X-rays and aseptic surgery existed, many traditional home remedies (like wrapping brown paper and vinegar for headaches) remained highly common. [✓][X]
38. The introduction of the polio vaccine in the 1950s. [✓][X]
39. Organ transplants, prosthetic joint replacements, and keyhole surgery. [✓][X]
40. The evolution of drug-resistant strains of bacteria (superbugs like MRSA) due to the over-prescription and overuse of antibiotics. [✓][X]
41. 1928; at St Mary's Hospital in London [✓][X]
42. A contaminating mould had grown on the dish and killed off the surrounding staphylococci bacteria. [✓][X]
43. He lacked the specialized chemical team and funding needed to isolate, concentrate, and stabilize the active compound. [✓][X]
44. That Alexander Fleming single-handedly developed penicillin into a mass-produced, usable medicine. [✓][X]
45. He found that penicillin became completely ineffective when mixed with blood in test tubes, causing him to doubt its value in living people. [✓][X]
46. Howard Florey and Ernst Chain; at Oxford University [✓][X]
47. As a skilled biochemist, he successfully grew the mould in his laboratory and extracted the active chemical compound. [✓][X]
48. It proved that penicillin was a highly effective antibiotic capable of killing deadly infections inside living bodies. [✓][X]
49. Penicillin was incredibly difficult to produce in large quantities, requiring massive volumes of grown mould to extract tiny amounts of active liquid. [✓][X]
50. Albert Alexander, a local policeman who developed septicaemia after being scratched by a rose bush. [✓][X]
51. The patient initially showed dramatic recovery, but died after the tiny supply of penicillin ran out, despite the doctors trying to recycle the drug from his urine. [✓][X]
52. British firms were fully occupied with the war effort, manufacturing essential military supplies and ammunition for World War II. [✓][X]
53. The United States [✓][X]
54. The entry of the United States into World War II, making the treatment of wounded soldiers a matter of urgent national priority. [✓][X]
55. Growing the mould in giant beer vats and deep-fermentation tanks. [✓][X]
56. Over 2.3 million doses, enough to treat all Allied casualties [✓][X]
57. He refused to patent the drug, declaring that it should be a free, lifesaving resource available to everyone in the world. [✓][X]
58. Dorothy Crowfoot Hodgkin [✓][X]
59. 1945 [✓][X]
60. Individual brilliance (Fleming/Florey/Chain), institutions (the US Government), science/technology (deep fermentation), and the catalyst of war (WWII). [✓][X]
61. 1950 [✓][X]
62. Because the government earned around £4 billion from tobacco tax, thousands of jobs depended on the industry, and there were ethical debates about limiting personal freedoms. [✓][X]
63. They were not clear enough, meaning tumors were often missed entirely or mistaken for other things like lung abscesses. [✓][X]
64. A CT scan uses dye and takes detailed 3D images, allowing doctors to spot tiny tumors when they are the size of a pea. [✓][X]
65. It shows how active the cancer cells are by tracking how much energy the tumor is using, helping doctors see if a treatment is working. [✓][X]
66. A flexible tube with a camera passed down the patient's airway, allowing doctors to see inside the lungs and collect cell samples. [✓][X]
67. It is the extraction and scientific analysis of a tiny sample of lung tissue to confirm if a tumor is indeed cancerous. [✓][X]
68. It uses powerful drugs (chemicals) to attack and kill cancer cells throughout the body, but it also damages healthy cells, causing severe side effects. [✓][X]
69. Using targeted, high-energy radiation beams to shrink and destroy localized cancer tumors. [✓][X]
70. Using specialized drugs to 'wake up' and boost the patient's own immune system so it can identify and destroy cancer cells. [✓][X]
71. Banning cigarette advertisements on television. [✓][X]
72. 2005 [✓][X]
73. They raised the legal age for buying tobacco from 16 to 18. [✓][X]
74. A ban on smoking in all enclosed workplaces, including pubs, cafés, restaurants, and offices. [✓][X]
75. A requirement that all tobacco products in shops must be removed from open display and kept hidden from sight. [✓][X]
76. By banning smoking in all cars carrying children under the age of 18. [✓][X]
77. Compulsory plain, olive-green packaging featuring prominent graphic health warnings and no brand logos. [✓][X]
78. Because the available tests are not accurate enough to outweigh the negative effects of screening, such as exposing healthy people to radiation. [✓][X]

Question 3: Similarity & Difference Across Eras (4 Marks)

P-F-C Formula · ~6 Mins

Explain one way in which ideas about the cause of illness in the Medieval period (c1250–c1500) were similar to ideas about the cause of illness in the Renaissance period (c1500–c1700).

🚀 **Level 1 Launchpad:** Starter: *"One way in which ideas about the cause of illness were similar was the continuing belief in..."*
Facts to Include: Persistence of miasma (bad air) and the Four Humours; despite Harvey and Vesalius disproving Galen on anatomy, ordinary physicians in 1665 still blamed the Great Plague on miasma and humours, just as in 1348.

One way in which ideas about the cause of illness were similar across both eras was...

In the Medieval period, people believed that...

Similarly, in the Renaissance period, (provide precise evidence from the 1665 Great Plague)...

This shows a strong continuity in medical thinking because...

⚡ **Level 2 (Grade 9) Comparison Rule:** You must explicitly refer to BOTH periods with equal factual depth. Do not spend all your time describing the Medieval period and leave the Renaissance to a single sentence!

Question 4: Causal Explanation (12 Marks)

3 P-E-E Paragraphs · ~18 Mins

Explain why there was rapid progress in surgical techniques in the period c1700–c1900.

You may use the following in your answer: • **James Simpson and chloroform** • **Joseph Lister and carbolic acid** *(You must also use information of your own.)*

🚀 **Level 1 Launchpad:** Paragraph 1: Anaesthetics (Simpson 1847) solved pain; Paragraph 2: Antiseptics (Lister 1867) solved infection; Paragraph 3 (Own Knowledge): Aseptic surgery (sterilization, rubber gloves) and ligatures solved blood loss.

PARAGRAPH 1: STIMULUS FACTOR 1 — THE CONQUEST OF PAIN (SIMPSON & CHLOROFORM)

One major reason for surgical progress was the overcoming of pain through anaesthetics...

In 1847, James Simpson discovered that chloroform...

This transformed surgery because doctors could perform deeper, more complex operations...

PARAGRAPH 2: STIMULUS FACTOR 2 — THE CONQUEST OF INFECTION (LISTER & CARBOLIC ACID)

A second critical factor was the reduction of post-operative infection...

Inspired by Pasteur's Germ Theory, Joseph Lister used carbolic acid in 1867 to...

Consequently, mortality rates in his surgical wards plummeted from...

PARAGRAPH 3: OWN KNOWLEDGE FACTOR — ASEPTIC SURGERY & LIGATURES

Furthermore, surgery advanced beyond antiseptics through the development of aseptic surgery...

For example, (detail steam sterilization, Halsted's rubber gloves, or surgical catgut)...

⚡ **Level 3 (Grade 9) Analytical Glue:** Explain how Simpson's discovery initially led to the 'Black Period' of surgery (higher infection from operating deeper) *until* Lister's antiseptics solved the sepsis crisis!

Question 5 / 6: Either/Or Statement Essay (16 Marks + 4 SPaG)

Balanced Evaluation · ~30 Mins

"Louis Pasteur's Germ Theory was the most significant turning point in the prevention of disease between c1700 and the present."

How far do you agree with this statement? Explain your answer.

You may use the following in your answer: • **The 1875 Public Health Act** • **Fleming, Florey and Chain's development of penicillin** *(You must also use information of your own.)*

Side A: Agree (Pasteur & Koch)
Disproved spontaneous generation; proved specific microbes cause specific diseases; unlocked scientific vaccines (rabies, anthrax) and modern hygiene.

Side B: Government Action
1875 Public Health Act made sanitation compulsory; clean water, sewers, and rubbish disposal saved far more working-class lives than laboratory science alone.

Side C: Modern Breakthroughs
Penicillin (1941) provided the first cure for internal bacteria; modern lifestyle campaigns (anti-smoking) and DNA (1953) target non-infectious causes.

INTRODUCTION & PARAGRAPH 1: AGREE WITH NAMED FACTOR (PASTEUR'S GERM THEORY)

To a significant extent, I agree that Louis Pasteur's Germ Theory was a profound turning point...

Prior to 1861, doctors believed in spontaneous generation, which meant...

By proving in his swan-neck flask experiments that germs caused decay, Pasteur enabled Robert Koch to...

Consequently, this transformed prevention by allowing scientists to develop targeted vaccines for...

PARAGRAPH 2: COUNTER-FACTOR — GOVERNMENT ACTION & PUBLIC HEALTH

However, it can be strongly argued that the 1875 Public Health Act was a greater practical turning point...

While Pasteur understood the cause in a laboratory, ordinary people were dying from filthy urban conditions...

By abandoning laissez-faire and legally forcing local councils to provide clean water and sewers...

Consequently, this administrative intervention saved millions from water-borne epidemics like cholera...

PARAGRAPH 3: MODERN ERA & OTHER FACTORS (PENICILLIN / DNA / PREVENTION)

Furthermore, in the 20th century, the nature of disease prevention shifted towards...

For example, while Fleming, Florey, and Chain's penicillin provided an unprecedented cure...

Modern prevention has increasingly focused on lifestyle campaigns and genetic understanding (DNA)...

CONCLUSION: CRITERIA-LED JUDGEMENT (WHY ONE FACTOR WAS DECISIVE)

In conclusion, while Pasteur's Germ Theory provided the indispensable scientific foundation for understanding disease...

The true turning point in national disease prevention was government public health intervention because...

SPaG Checklist (+4 Marks): [ ] Accurate spelling of key terms (Pasteur, bacterium/bacteria, antiseptic, laissez-faire); [ ] Sustained analytical tone throughout.

Question 3: Similarity & Difference Practice (4 Marks)

Cross-Period Comparative Matrix

Explain one way in which the government's response to the Great Plague (1665) was different to the response to the Black Death (1348).

⏱ Comparative Fact Vault (AO1)

1348 Black Death: Government action was minimal; King Edward III ordered street cleaning; local authorities were powerless; relied on church prayers.

1665 Great Plague: Mayor of London enforced strict quarantine; searchers identified infected; red crosses painted on doors; watchmen guarded houses; mass graves.

Difference Connectives

- *"In contrast to 1348, authorities in 1665..."*
- *"While medieval response was largely fatalistic, Renaissance response..."*

🎯 **Level 2 Standard:** Focus strictly on GOVERNMENT/AUTHORITY action in both periods.

One way in which the government's response was different was the degree of organized quarantine...

In 1348, during the Black Death, authorities took very little organized action because...

In contrast, during the Great Plague of 1665, the Mayor of London enforced strict regulations such as...

This shows that by 1665, authorities took a far more active, systematic approach to containment...

Section B · Turn page for Question 4 Dual Track

Question 4: Causal Explanation Practice (12 Marks)

Penicillin Breakthrough

Explain why there was rapid progress in the development and mass production of penicillin in the years c1938–c1945.

Stimulus: • **Howard Florey and Ernst Chain** • **US Government funding and the Second World War**

⏱ Causal Factor Vault (AO1)

Florey & Chain (1938–41): Purified Fleming's 1928 mould; tested on 8 mice (1940); human trial on Albert Alexander (1941) proved it killed infection in blood.

US War Funding (1941–44): British factories made bombs; US government funded deep fermentation tanks; enough penicillin to treat all Allied D-Day wounded (June 1944).

Own Knowledge: Cantaloupe melon mould strain found in Peoria, Illinois; corn steep liquor boosted yield 20x.

🎯 **Level 3 Standard:** Explicitly show how the pressure of WAR overcame the financial and industrial barriers to mass production.

PARA 1: FLOREY & CHAIN'S SCIENTIFIC BREAKTHROUGH

The primary scientific catalyst for progress was the biochemical work of Florey and Chain...

This was critical because while Fleming discovered penicillin accidentally in 1928...

PARA 2: SECOND WORLD WAR & US STATE INTERVENTION

However, scientific discovery could not achieve mass production without wartime urgency...

Recognizing its battlefield value, the US government poured millions of dollars into...

PARA 3: INDUSTRIAL TECHNOLOGY & AGRICULTURAL BREAKTHROUGHS

Furthermore, mass production was unlocked by industrial innovations such as...

 Section B · Turn page for Questions 5/6 Dual Track

Question 5 / 6: Statement Essay Practice (16+4 Marks)

Four Humours c1250–c1700

"The Theory of the Four Humours was the most significant foundation of medical knowledge between c1250 and c1700."
How far do you agree? Explain your answer.

Stimulus: **• The enduring influence of Galen** **• Traditional herbal remedies**

⏱ Continuity vs Change Matrix

Four Humours Dominance: Endorsed by Church; Galen's texts taught in universities; phlebotomy, purging, regimen sanitatis; used to explain 1348 and 1665 plagues.

Renaissance Challenges: Vesalius (1543) proved 300+ errors in Galen; Harvey (1628) proved circulation; Sydenham (1676) grouped diseases by symptoms not individual humours.

🎯 **Level 4 Evaluation:** Distinguish between theoretical breakthroughs in universities and the continuity of humeral treatments among ordinary people.

PARA 1: THE ENDURING DOMINANCE OF THE FOUR HUMOURS

On the one hand, the Four Humours was unquestionably the dominant foundation of medicine...

Supported by the Catholic Church, Galen's humeral theories were taught as indisputable dogma...

PARA 2: RENAISSANCE CHALLENGES TO HUMERAL THEORY

However, between c1500 and c1700, scientific empirical observation began to dismantle humeral ideas...

Vesalius proved Galen wrong on anatomy in 1543, and William Harvey proved...

⏱ Alternative Factors Toolkit

Herbal Remedies: Practical folklore medicine passed down by women/wise-women; honey, mint, willow bark; independent of Greek theory.

Sydenham & Observation: Observed epidemics as external clinical entities rather than internal balance of humours.

⚡ **Grade 9 Judgement:** Conclude that while anatomy changed rapidly, actual treatment remained fundamentally humeral until the 18th century!

PARA 3: TRADITIONAL HERBAL & ALTERNATIVE APPROACHES

Furthermore, for the vast majority of ordinary people, medical knowledge was founded on...

Herbal remedies and community care provided by apothecaries and local women...

CONCLUSION: SUPPORTED FINAL JUDGEMENT

In conclusion, I agree to a large extent that the Four Humours was the most significant foundation...

Although anatomical discoveries by Vesalius and Harvey laid the groundwork for modern science...

In terms of everyday clinical treatment, humeral bleeding and purging remained unchallenged until c1700...

SECTION B: MEDICINE IN BRITAIN, C1250–PRESENT

Round 3: The Planning Engine Room · Thematic Causal & Synoptic Deconstruction

ROUND 3: ENGINE ROOM

1. Cross-Period Comparison Engine (Q3 Practice):

Comparison Question	Period 1 Specific Evidence	Period 2 Specific Evidence
Treating Epidemics: Black Death vs Great Plague	1348: Flagellation, church prayers, carrying herbs, vinegar.	1665: Watchmen locked doors, searchers, burning barrels of pitch, plague pits.
Hospital Care: Medieval vs 19th Century	c1300: Monasteries (St Bartholomew's); focus on spiritual rest; no doctors.	c1860: Nightingale pavilion wards; trained nurses; antiseptic surgery.

2. Overarching Thematic Factors Matrix (Q4 & Q5/Q6):

Core Factor	Key Historical Examples Across 750 Years	Direct Analytical Impact
Government Action	1875 Public Health Act, compulsory smallpox vaccination (1853), NHS (1948)	Enforces sanitation and equal access beyond market forces.
Science & Tech	Printing press (1476), Microscope, Swan-neck flasks, X-rays, DNA (1953)	Provides empirical proof that overturns dogma.
Key Individuals	Vesalius, Harvey, Jenner, Simpson, Lister, Pasteur, Koch, Snow, Fleming	Acts as intellectual pioneers driving revolutionary change.

📋 **PRE-FLIGHT AUDIT CHECKLIST:** [ ] Did I compare BOTH eras with factual detail in Q3? [ ] Did I explain THREE distinct causes for Q4? [ ] Is my Q5/Q6 essay structured around criteria-led evaluation?

Section B · Execute continuous prose on Page 23

SECTION B: MEDICINE IN BRITAIN, C1250–PRESENT

Round 3: The Exam Pitch · Authentic Edexcel Section B Simulation (36+4 Marks · 55 Mins)

TIMED EXAM PITCH

AO1 Knowledge: Precise names, dates, individuals, and acts across all four eras. **AO2 Analysis:** Sustained explanation of similarity, causation, and relative importance.

Question 3 (4 Marks): *One way in which hospital care in the 19th century differed from medieval hospitals was...*

Question 4 (12 Marks): *Explain why penicillin was mass-produced rapidly c1938–45: ___________________________*

Question 5 / 6 Essay (16+4 Marks): *"____"*

⚠️ EXAMINER TRAP DOORS & 100% SPECIFICATION BANK

Section B: Medicine in Britain, c1250–present · Pearson Edexcel GCSE History

100% Spec Guarantee

🚫 TOP 3 FATAL EXAMINER TRAPS TO AVOID FOR SECTION B

• The One-Period Comparison Trap
In Question 3, you cannot get above Level 1 if you only talk about one era. You must give balanced, explicit factual evidence for BOTH periods!

• The Overnight Miracle Trap
Never describe scientific breakthroughs (Vesalius, Harvey, Jenner, Pasteur) as instant cures. In reality, ordinary treatments took decades or centuries to catch up!

• The Storytelling Narrative Trap
Question 4 and 5/6 are causal and evaluative essays, NOT stories. Never just list what Jenner or Snow did; explicitly explain *why* it changed outcomes.

📚 SPECIFICATION PRACTICE BANK: 100% THEMATIC CURRICULUM COVERAGE

Every remaining thematic specification bullet point tested below

1. Q3 Similarity (4m): Explain one way in which the training of physicians in the Medieval period was similar to the training of physicians in the Renaissance.

2. Q4 Explain Why (12m): Explain why government attitudes to public health changed in the period c1848–c1875. (Edwin Chadwick, John Snow).

3. Q5/Q6 Essay (16+4m): 'The founding of the National Health Service in 1948 was the most significant breakthrough in medical care between c1900 and the present.' How far do you agree?

4. Q3 Difference (4m): Explain one way in which treatments for the Black Death (1348) were different to treatments for Cholera (1854).

5. Q4 Explain Why (12m): Explain why understanding the causes of disease advanced rapidly after 1900. (Watson & Crick/DNA, lifestyle research into lung cancer).

⚡ **750-YEAR SYNOPTIC TITANS:** Hippocrates & Galen (Humours) → Vesalius & Harvey (Anatomy) → Sydenham (Observation) → Jenner (Vaccines) → Simpson (Chloroform) → Lister (Antiseptics) → Pasteur & Koch (Germs) → Snow & Chadwick (Public Health) → Fleming, Florey & Chain (Antibiotics) → Watson, Crick & Franklin (DNA).